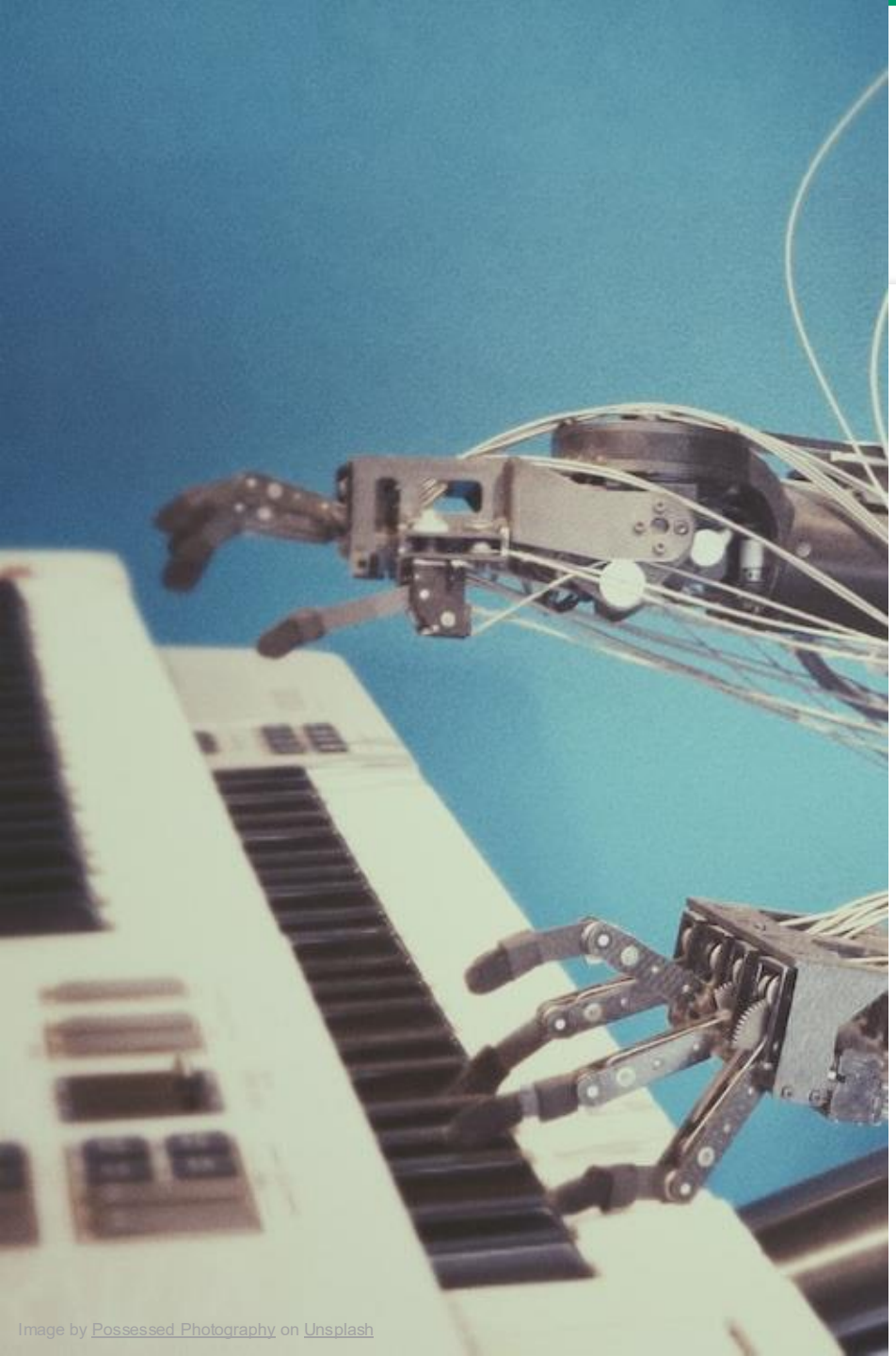


Applied Artificial Intelligence
05 - AI Lifecycle: Concept Drift

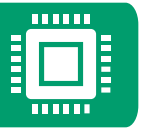
Univ.-Prof. Dr-Ing. habil. Niklas Kühl
www.niklas.xyz

University of Bayreuth

TUM School of Management

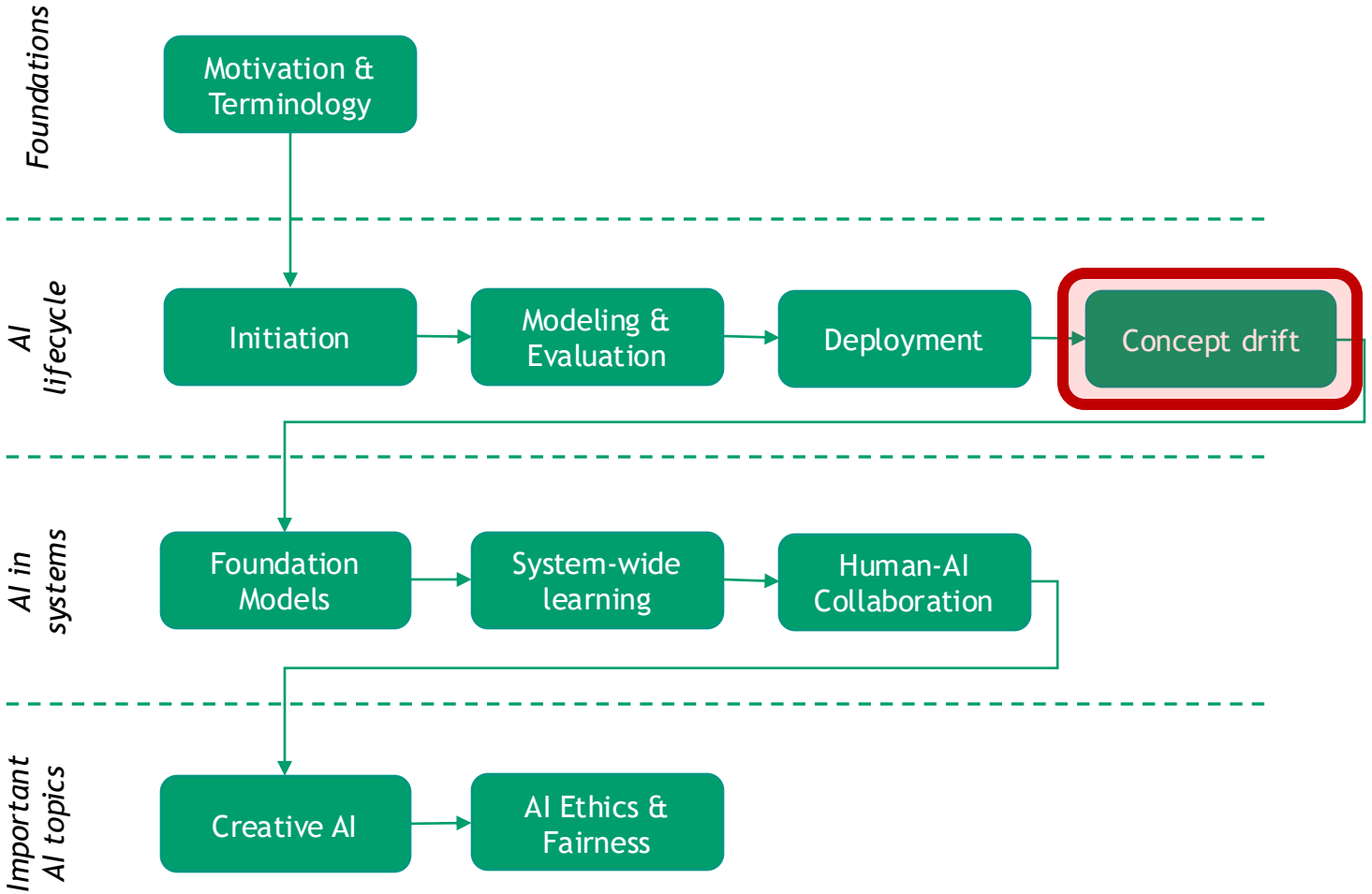
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Repetition

Training vs. deployment 	You learned that model training and testing come before operational deployment, involving iterative fine-tuning.
AI autonomy levels 	You discovered that AI applications can range from making predictions to offering recommendations and even taking direct actions.
Challenges of ML deployment 	You gained insights into dealing with multilingual support, harnessing the power of parallel GPUs, and managing unpredictable costs.
Deployment decisions 	You learned to choose between Monolithic and Microservices, and various deployment options, including On-Premise, IaaS, PaaS, FaaS, and SaaS.
Platforms for ML deployment 	You explored tools like Amazon SageMaker, Kubernetes, Cloud Foundry, and holistic MLaaS solutions.
Concept of MLOps 	You understood that MLOps is a comprehensive paradigm that involves a collaborative team, all working together to streamline the AI deployment process.

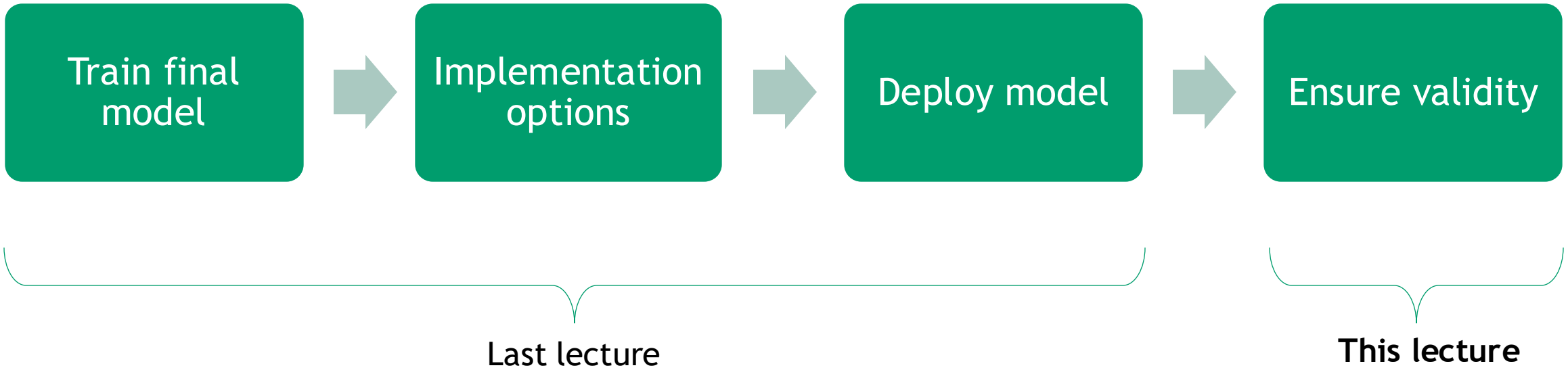
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Organizational
The story of the lecture



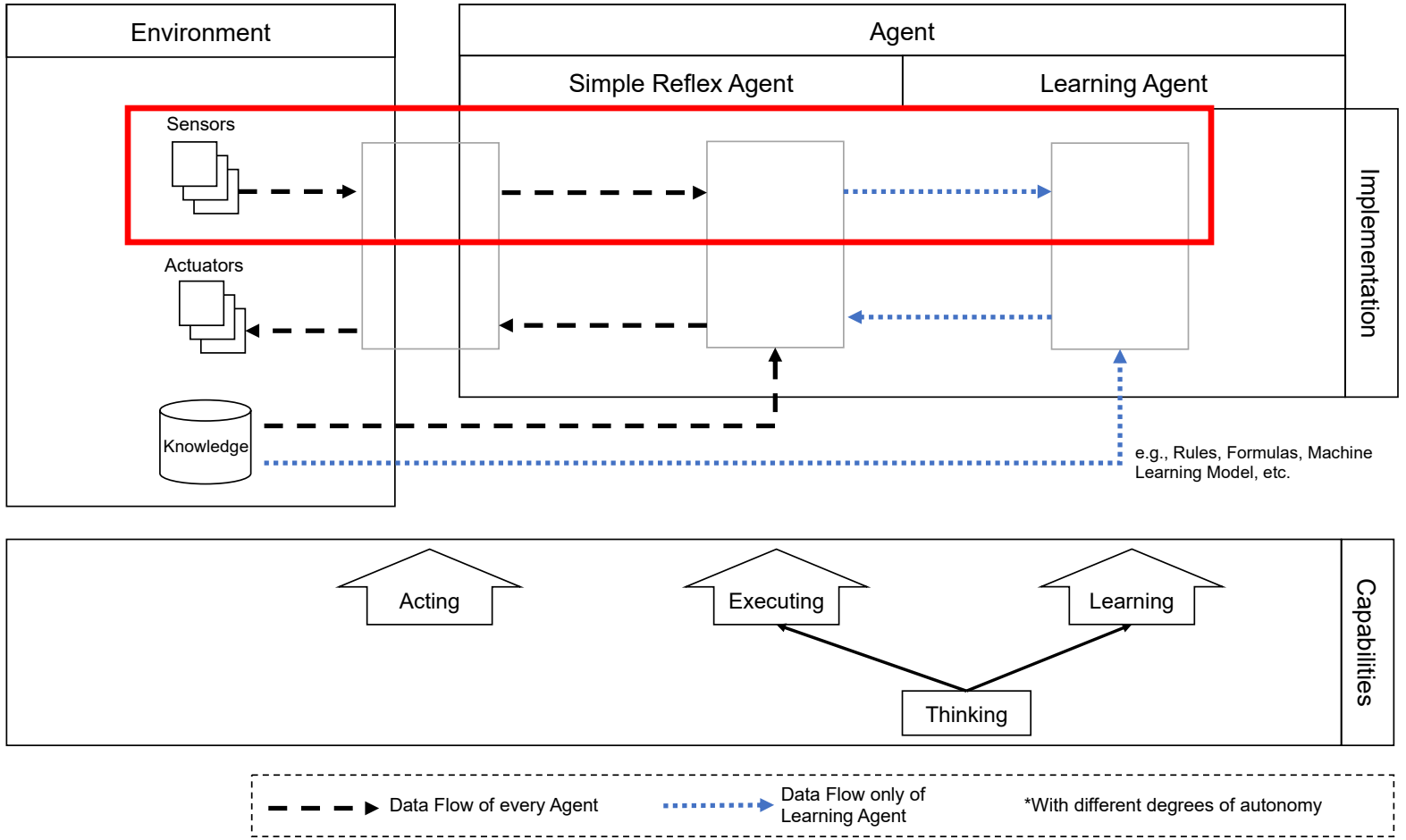
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Process



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Positioning of the “concept drift” within an intelligent agent’s architecture



Kühl et al. (2022), Artificial intelligence and machine learning [1]

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Objectives
What are the learning goals of this lecture?

EXPLORE

Learn about the term concept drift



UNDERSTAND

Understand the importance of continuous monitoring of deployed AI solutions



INTENSIFY

Familiarize with concept drift detection



APPLY

Be able to adapt prediction models



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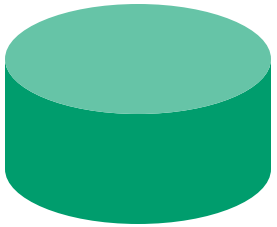
1. Introduction
2. Theoretical foundations of concept drift & adaption mechanism
3. Two examples for concept drift detection algorithms
4. Real-world examples for concept drift handling

Introduction
How to keep machine learning (microservices) correct?



- changing context or data over time
- own action taken on machine learning services

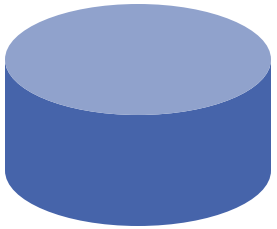




Training / test environment

ML model is based on training environment

Different data distribution



Productive environment

Continuous data stream

ML

Deployed Machine Learning Model

Image source DALL-E

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Introduction
Data streams may change in various application domains

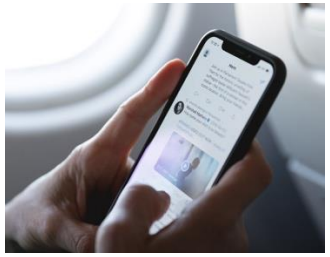
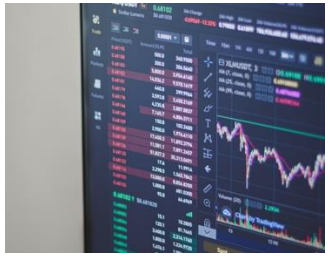
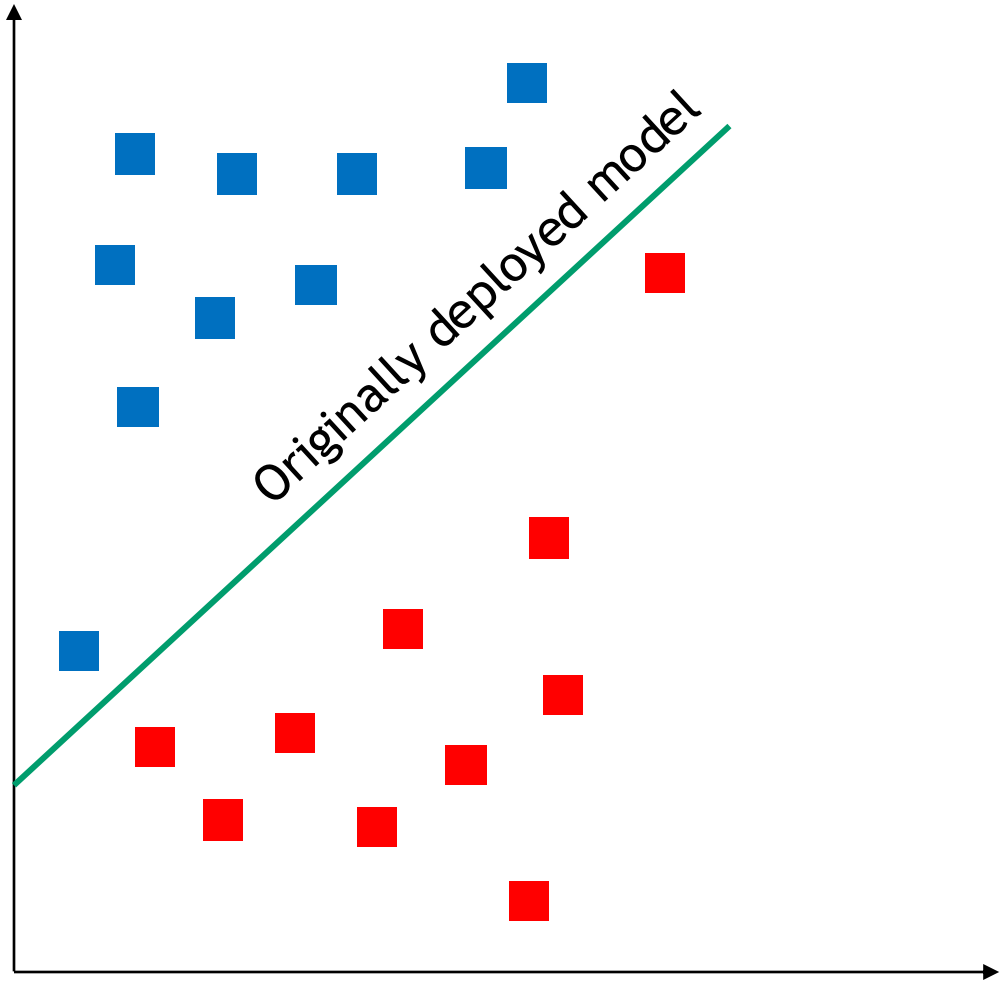
Changing context/concept of data	Machine learning	Result
 **Sentiment prediction:** Different communication pattern on Twitter (e.g. doubling of characters)	⇨	?
 **Prediction of prices:** Policy changes in electricity markets	⇨	?
 **Prediction of downtimes:** Change of machine parameters in industrial context	⇨	?

Image source https://unsplash.com, free licence

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Introduction

Ongoing validity of machine learning models can be ensured with model adaptations



Constant monitoring

… of data stream with dedicated change detection module (e.g. based on statistical properties)

Change detection

system decides whether model adaptation or retraining is necessary

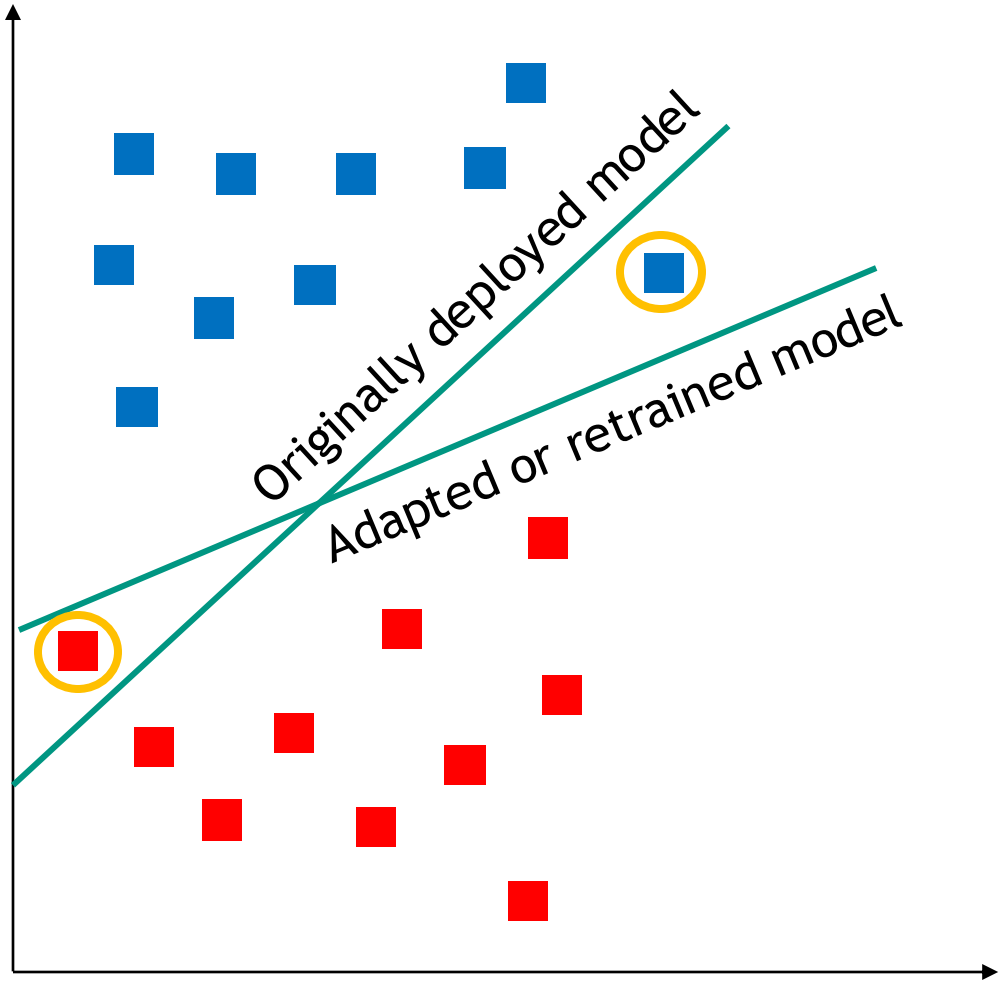
Adaptation: Machine learning model is slightly adapted based on the new data

Retraining: Machine learning model is completely retrained from scratch

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Introduction

Ongoing validity of machine learning models can be ensured with model adaptations



Constant monitoring

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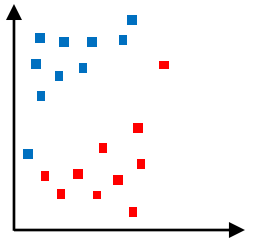
1. Introduction
2. Theoretical foundations of concept drift & adaption mechanism
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Theoretical foundations of concept drift & adaption mechanism

Concept drift describes the phenomenon of changing data in computer science

Definition of **concept:**

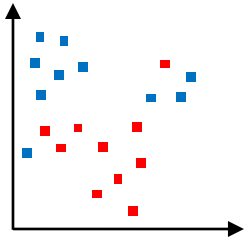
$$\text{Concept} = P(X, y)$$



Problem: In the real world, concepts are not stable but change with time

Definition of **concept drift** between two time points:

$$P_{t0}(X, y) \neq P_{t1}(X, y)$$



Closely related to **streaming data** with time stamps (e.g., social media data)

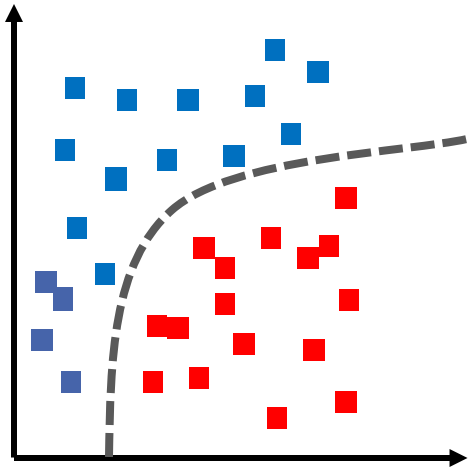
Learning algorithm has access to data with time stamps prior to t, but needs to be applied to data elements with subsequent time stamps

Source: Webb et al. (2016), Characterizing Concept Drift

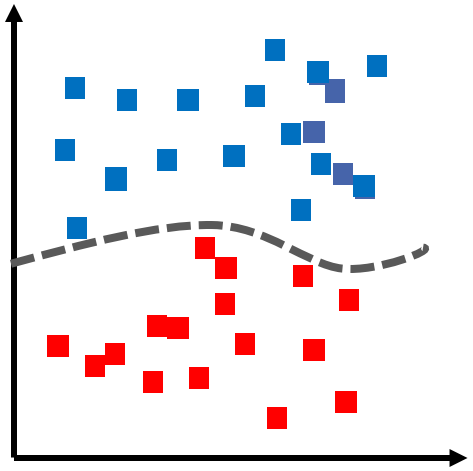
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Theoretical foundations of concept drift & adaption mechanism

Different types of concept drift exists

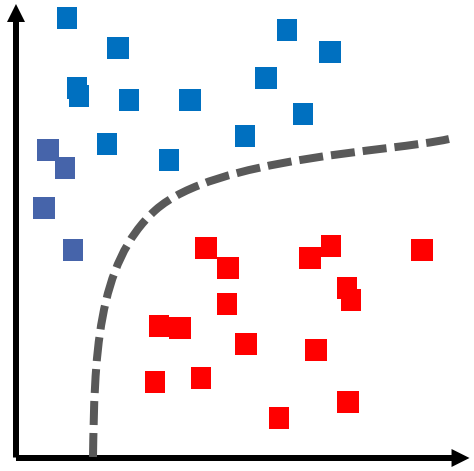


Original data



Real concept drift

changes in $P(y|X)$, $P(X)$ might stay constant or not



Virtual drift

$P(X)$ changes without affecting $P(y|X)$

Source: Gama et al. (2014), A survey on concept drift adaptation

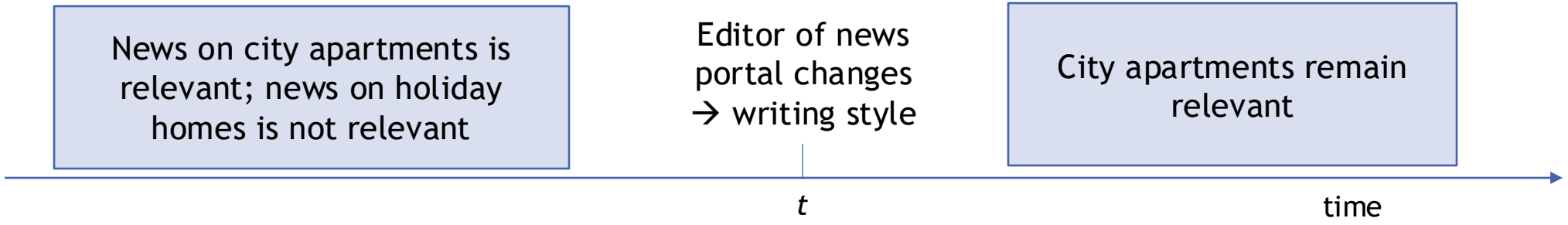
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Theoretical foundations of concept drift & adaption mechanism
Example: News classification

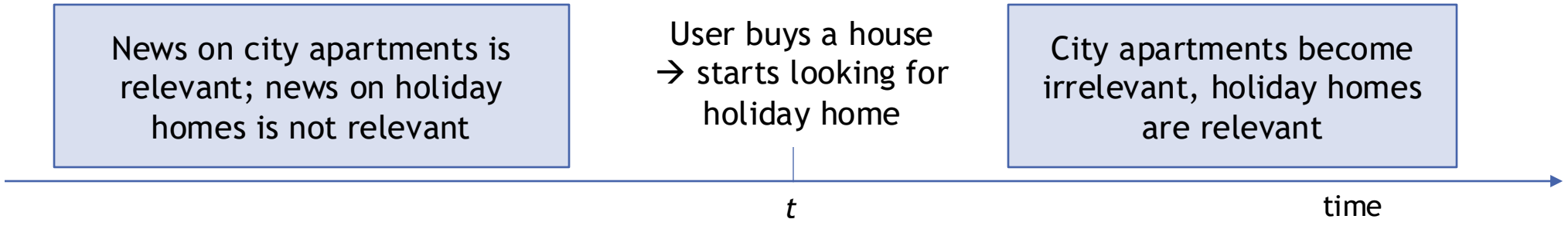
Task:
Classification of online news stream of articles on real estate as relevant or not relevant for a given user

Scenario: User is looking for a new apartment

Virtual drift:

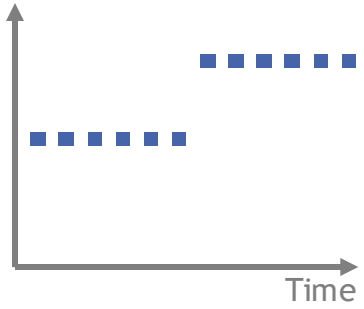


Real concept drift:



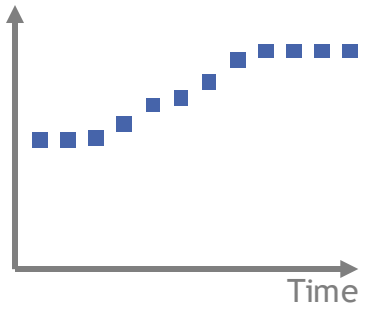
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Theoretical foundations of concept drift & adaption mechanism
Data changes over time in various ways



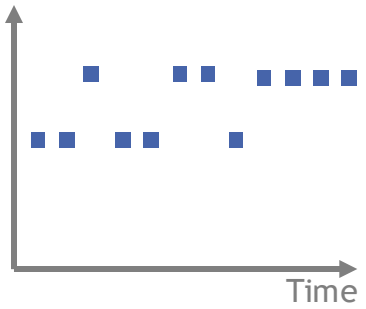
Sudden changes

Exchange of a sensor in IOT setting



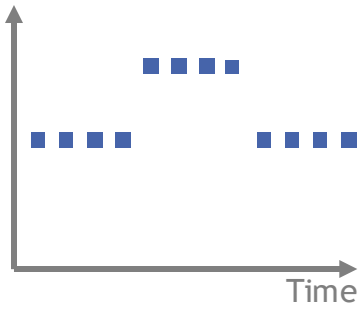
Slow changes

Sensor that is slowly degrading, becomes less accurate



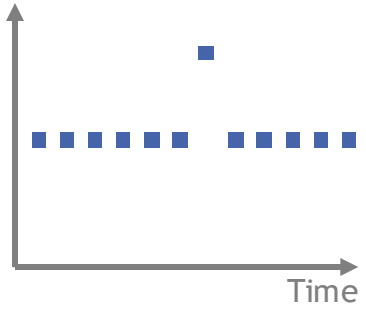
Gradual change

User behavior, user interested in finance, later in sports, but keeps looking back at finance



Reoccurring concepts

Seasonal pattern for sales forecasting



Outliers

Difficulties to not misclassify outliers as concept drift; Otherwise, danger of false adaptation

Source: Zliobaite (2010), Learning under Concept Drift

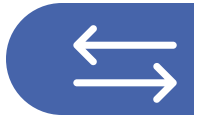
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Theoretical foundations of concept drift & adaption mechanism

Challenges for deployed machine learning services in concept drift environments



Detect
concept drift as soon as possible



Adapt
the model quickly if needed



Distinguish
drifts from noise; robustness to noise



Recognize
recurring contexts (e.g. seasons)



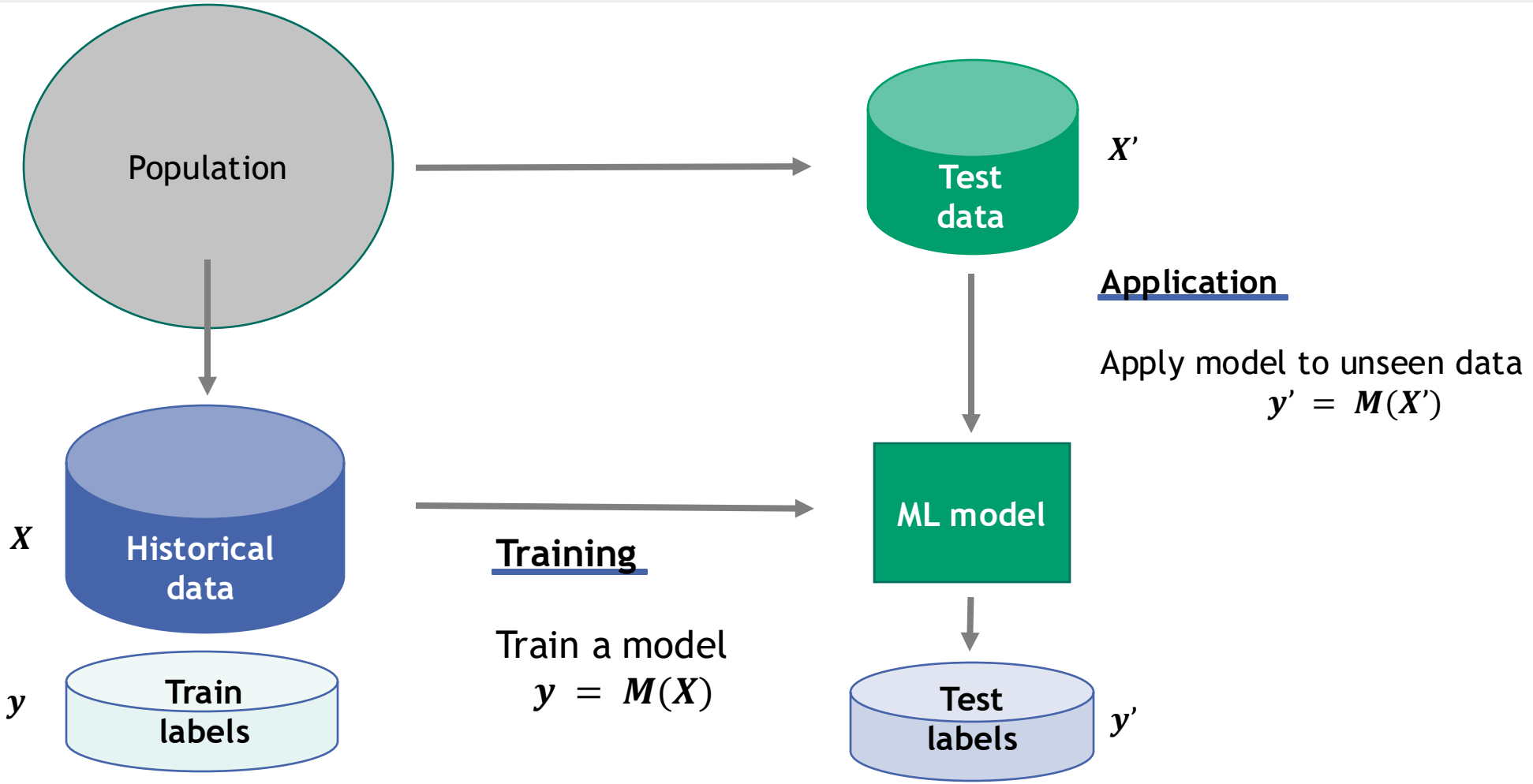
Operate
in less than example arrival time and with limited storage available

Source: Gama et al. (2014), A survey on concept drift adaptation Tsymbal (2004), The problem of concept drift: definitions and related work

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Theoretical foundations of concept drift & adaption mechanism

Recap: Supervised learning

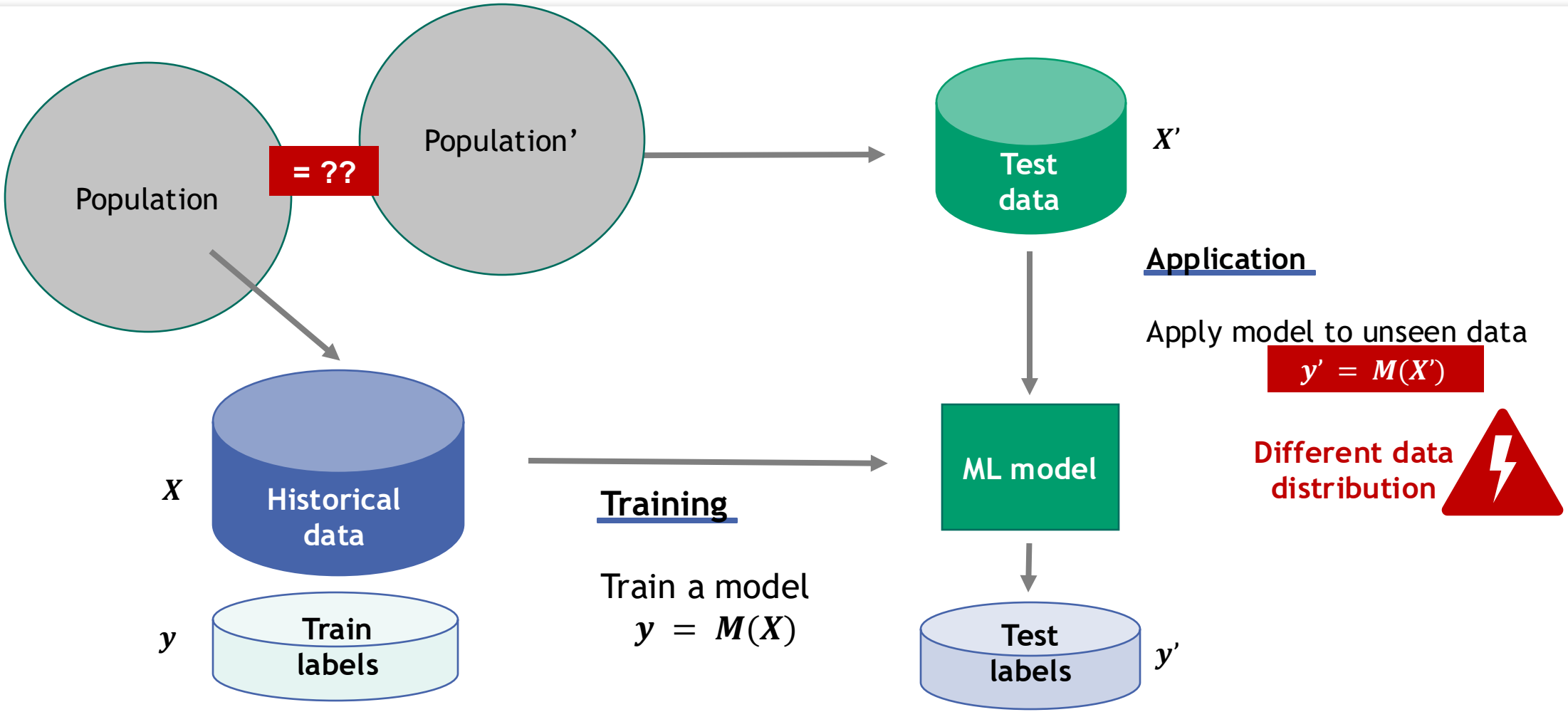


Source: Bifet et al. (2010), Handling Concept Drift

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Recap: Supervised learning

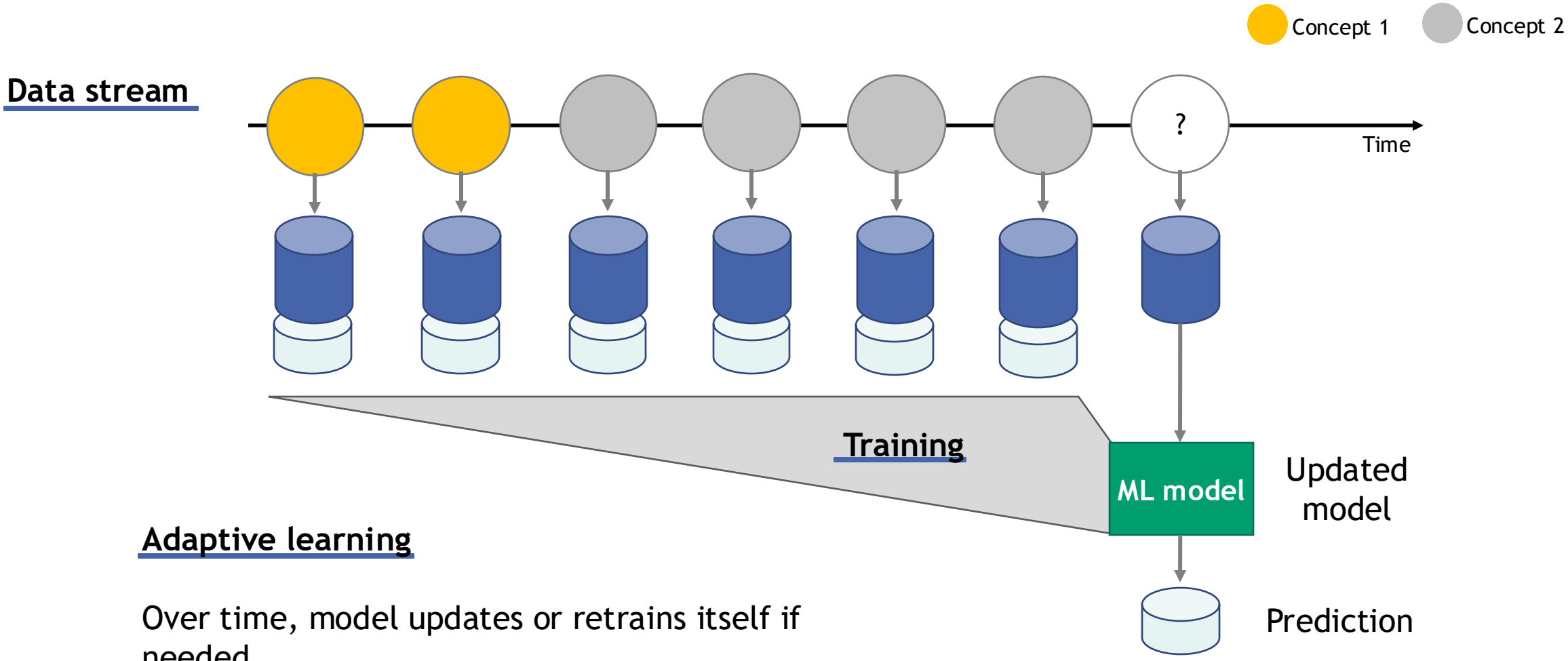


Source: Bifet et al. (2010), Handling Concept Drift

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Theoretical foundations of concept drift & adaption mechanism

We have to consider the machine learning model in a data stream / online setting



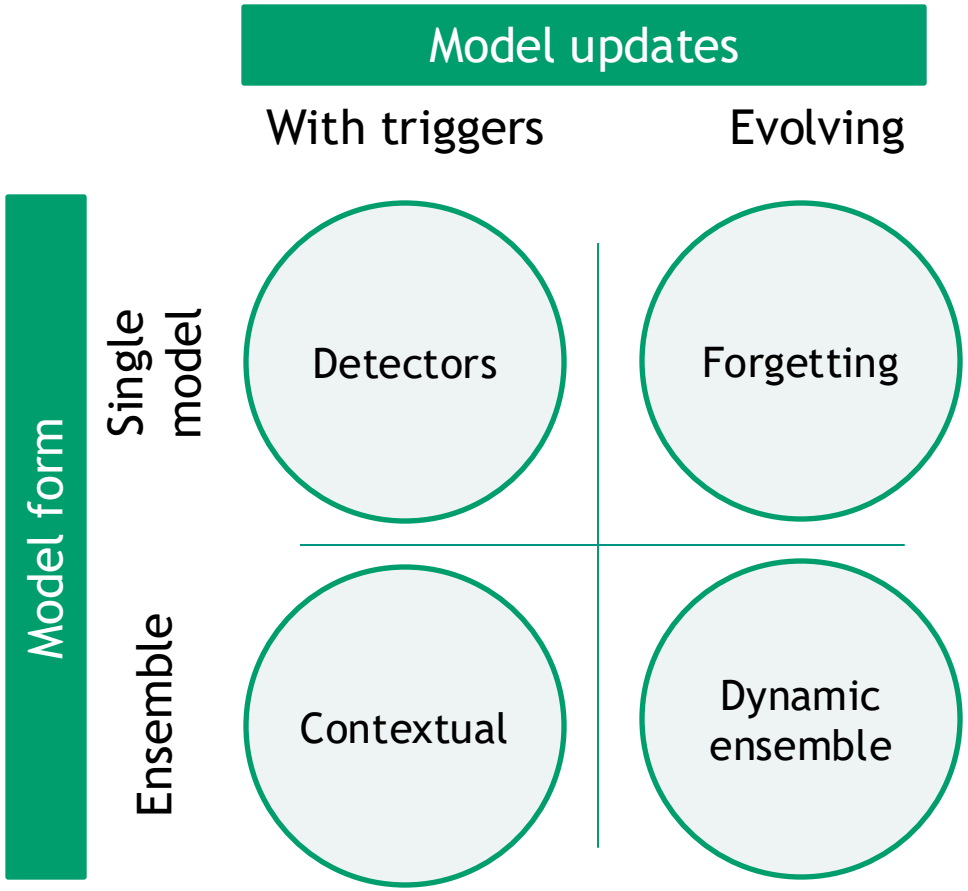
Adaptive learning

Over time, model updates or retrains itself if needed

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Theoretical foundations of concept drift & adaption mechanism

Different adaptive learning strategies are available

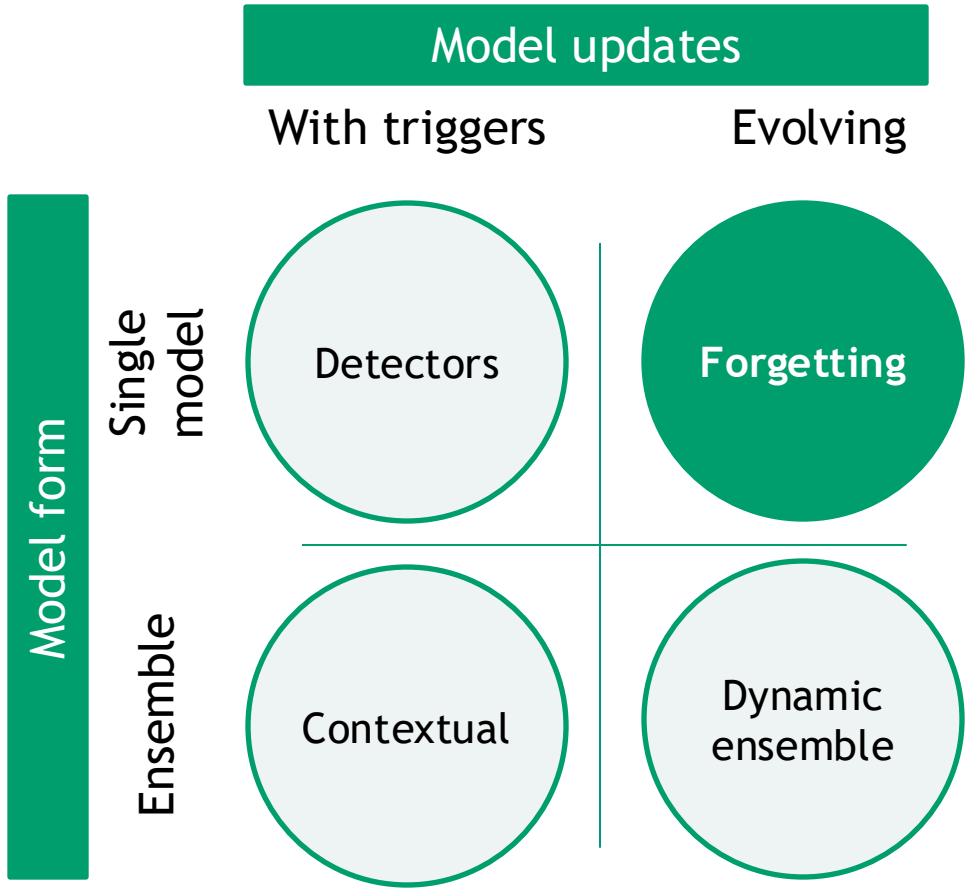


Source: Zliobaite (2010), Learning under Concept Drift

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Theoretical foundations of concept drift & adaption mechanism

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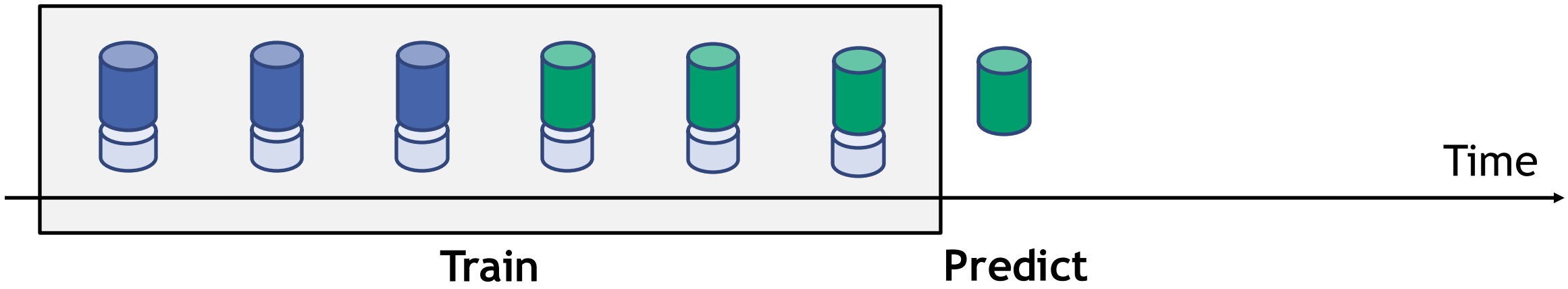


Source: Zliobaite (2010), Learning under Concept Drift

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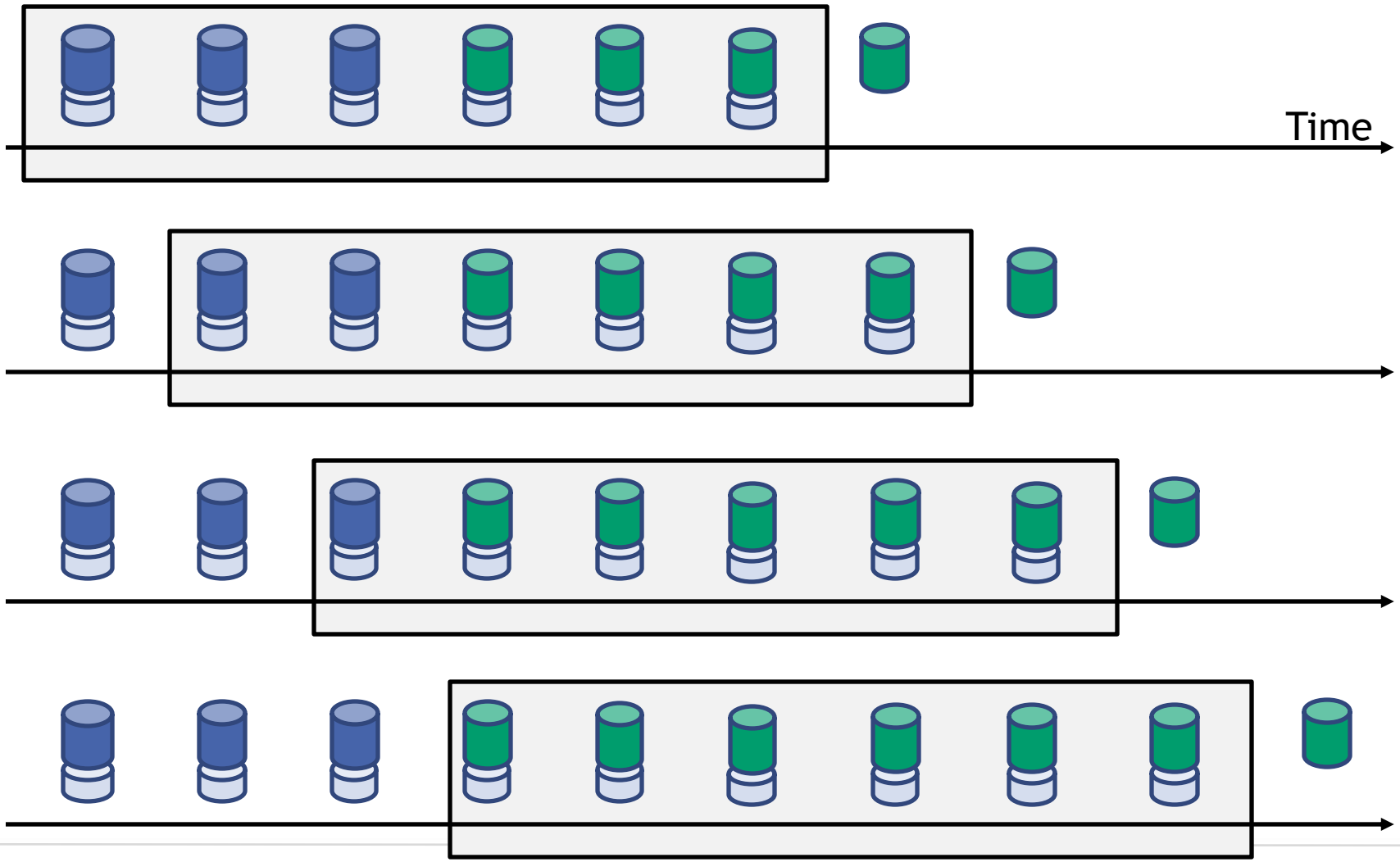
A fixed training window ensures forgetting



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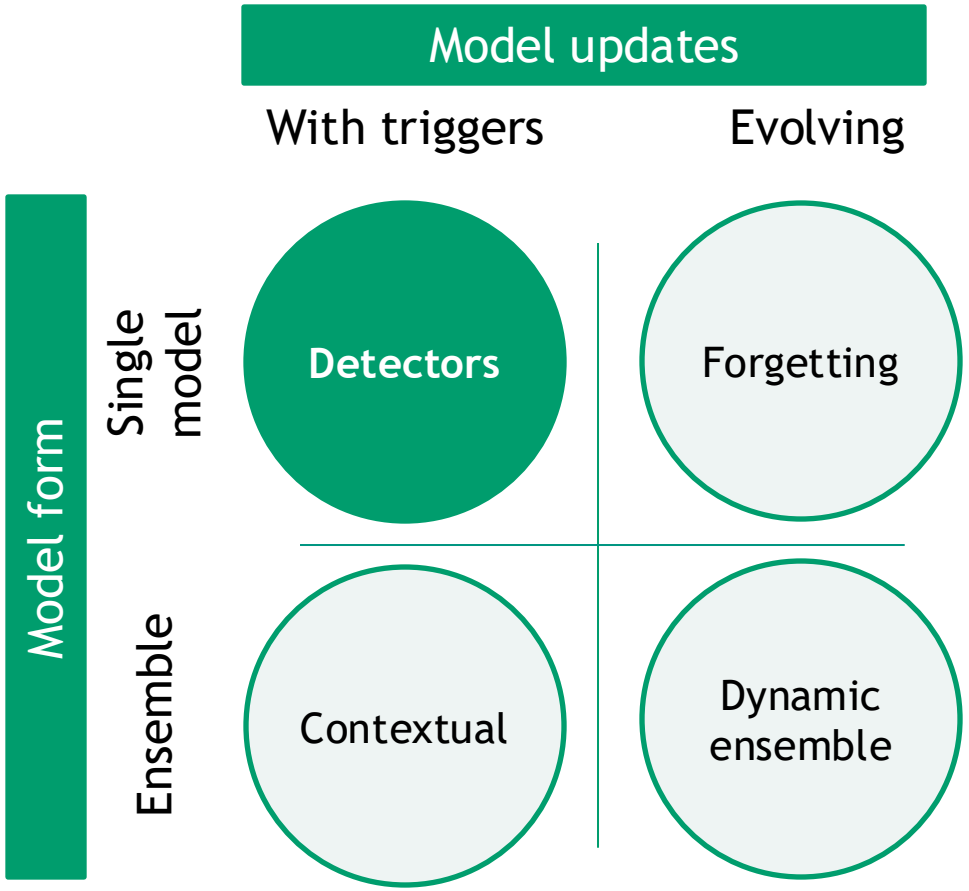
Theoretical foundations of concept drift & adaption mechanism

A fixed training window ensures forgetting



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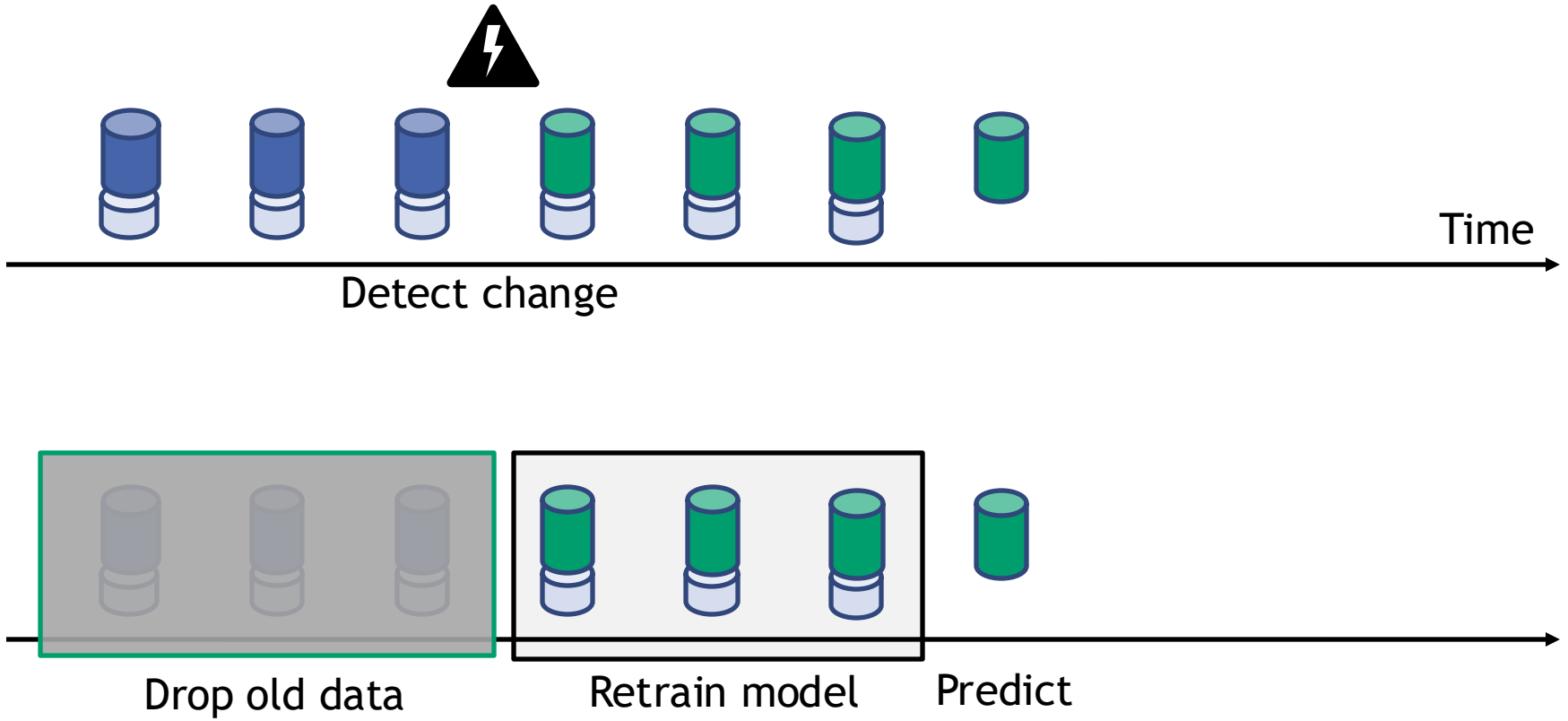
Theoretical foundations of concept drift & adaption mechanism
Different adaptive learning strategies are available



Source: Zliobaite (2010), Learning under Concept Drift

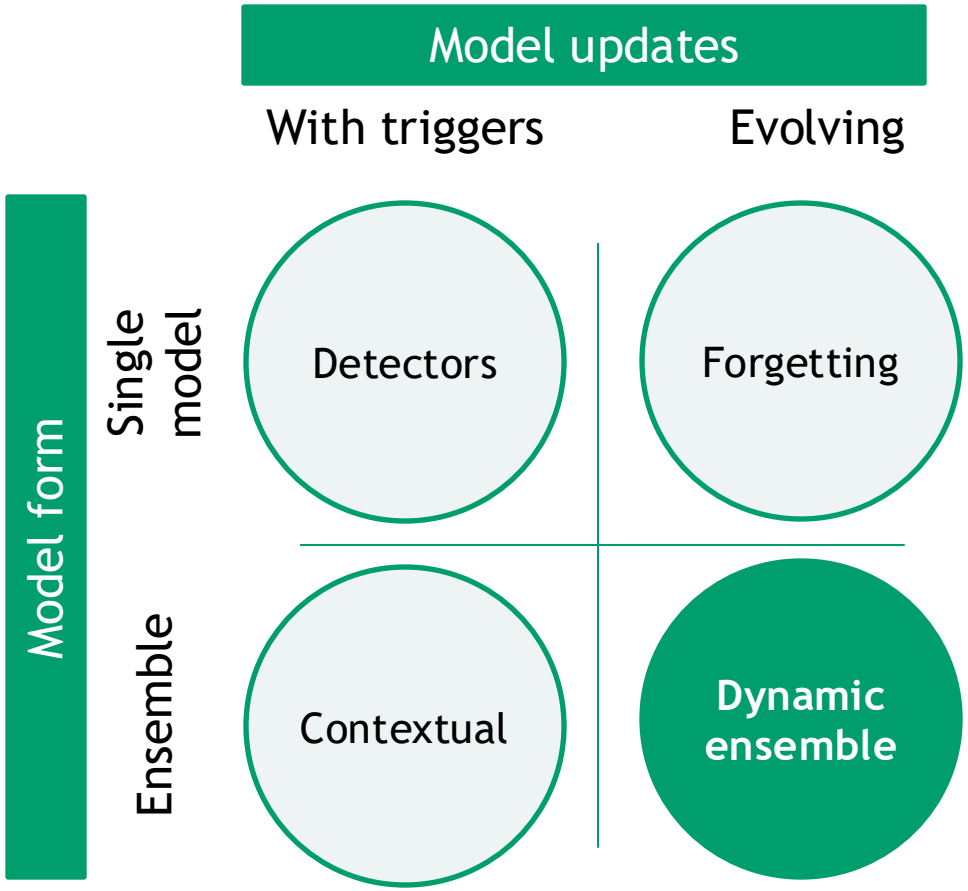
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Theoretical foundations of concept drift & adaption mechanism
Detection of change leads to deletion of old data



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Theoretical foundations of concept drift & adaption mechanism
Different adaptive learning strategies are available

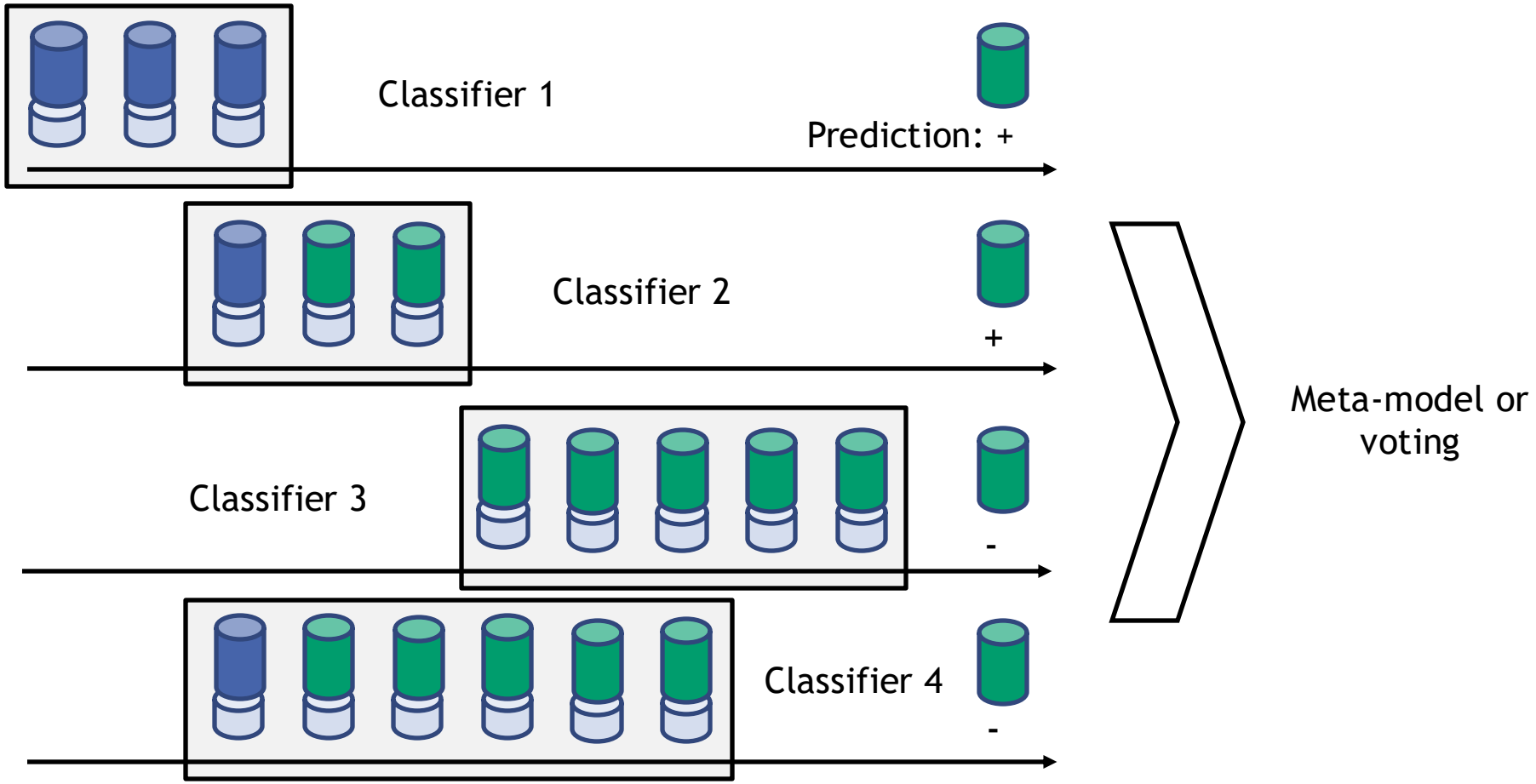


Source: Zliobaite (2010), Learning under Concept Drift

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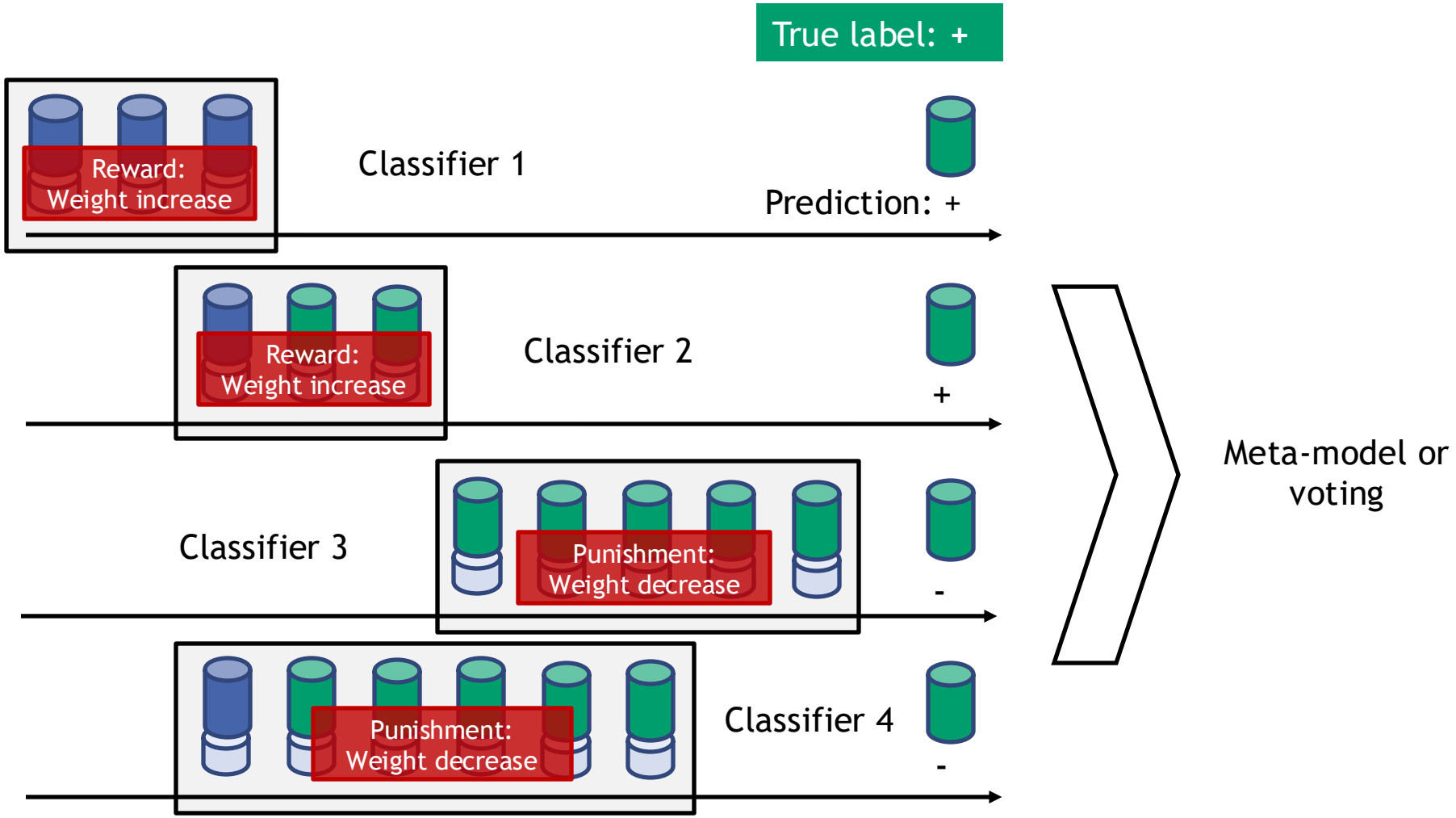
Theoretical foundations of concept drift & adaption mechanism

Dynamic ensembles consider several ML models in parallel



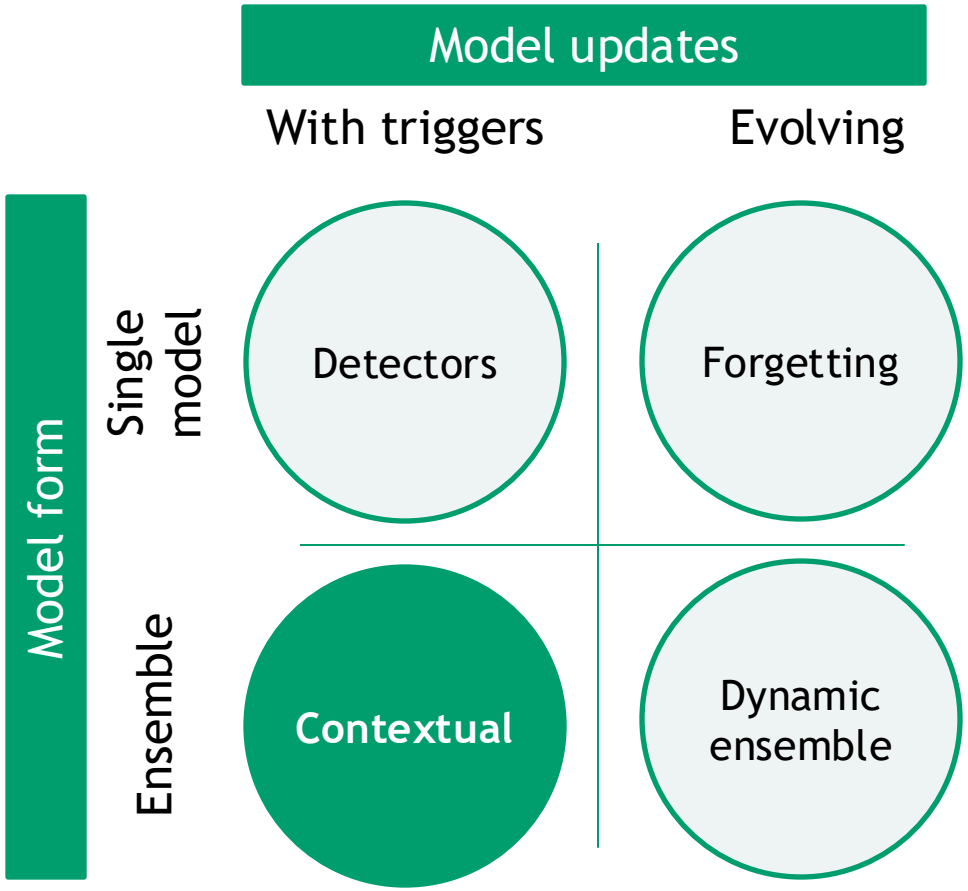
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Theoretical foundations of concept drift & adaption mechanism
Dynamic ensembles consider several ML models in parallel



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Theoretical foundations of concept drift & adaption mechanism
Different adaptive learning strategies are available

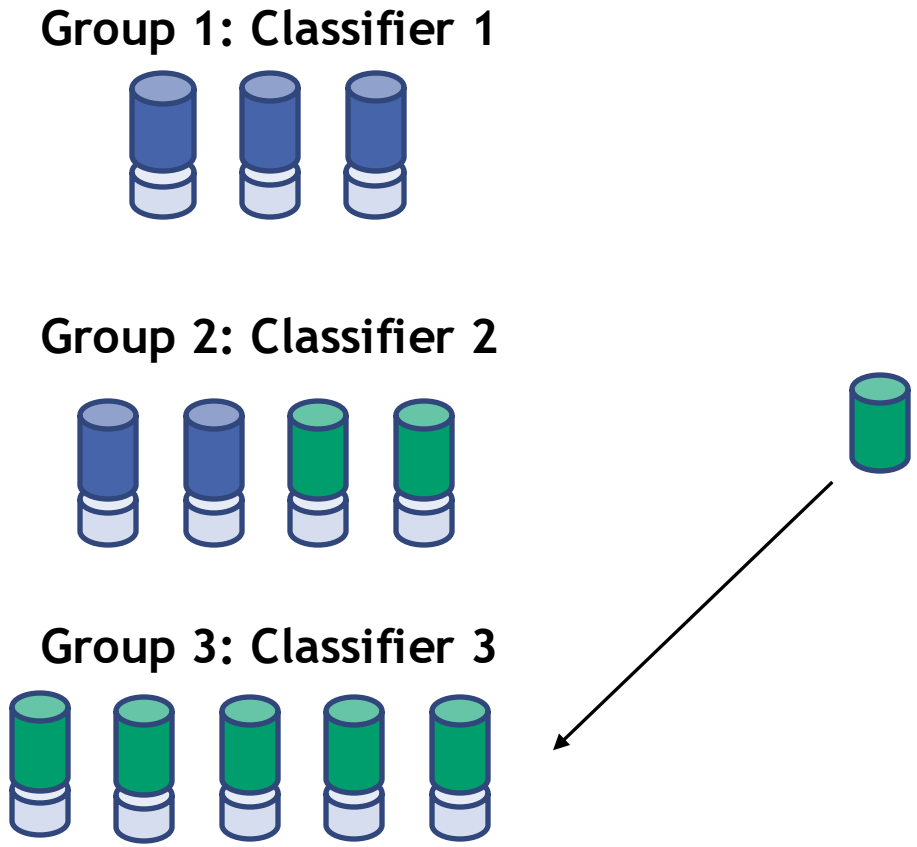


Source: Zliobaite (2010), Learning under Concept Drift

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Theoretical foundations of concept drift & adaption mechanism

Contextual approaches start by identifying the group affiliation of a new data instance



Train

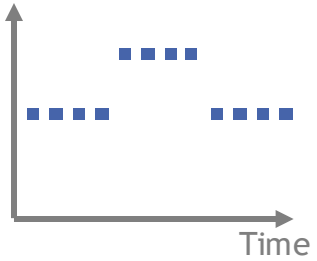
Partition the training data into several groups and build separate models for each group

Predict

New instance is assigned to one group and corresponding model is applied

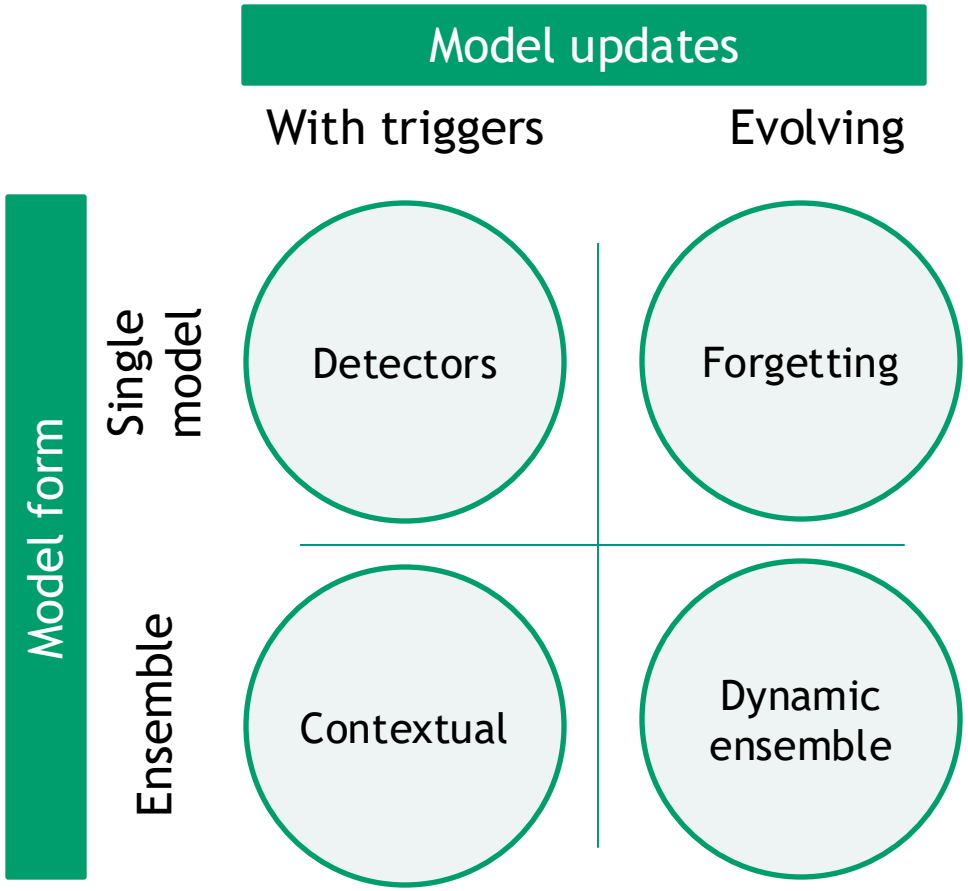
Use case

Especially suited for reoccurring concepts (e.g. seasonal pattern in sales forecasts)



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Theoretical foundations of concept drift & adaption mechanism
Different adaptive learning strategies are available



Source: Zliobaite (2010), Learning under Concept Drift

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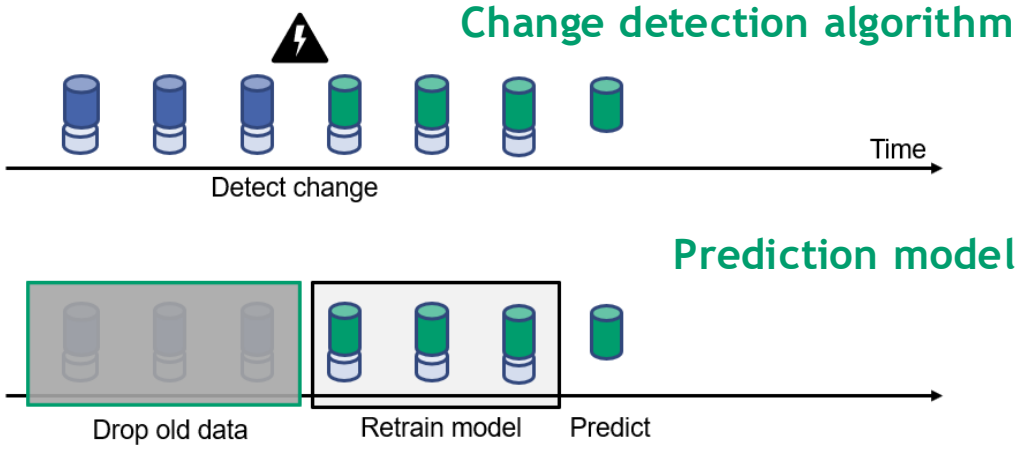


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Two examples for concept drift detection algorithms
Wide variety of change detection / adaptation approaches available

Informed methods Next! (Detector-based methods)

Explicit change detection



Change detection usually works on prediction error:

- Sequential Analysis: *Page-Hinkley*
- Monitoring two distributions: *ADWIN*

Blind methods (Forgetting-based methods)

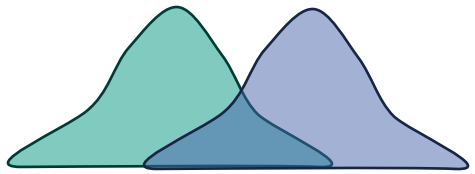
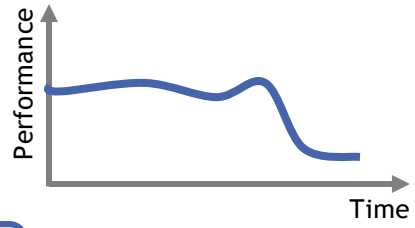
Model which adapt incrementally or are frequently retrained



- Incremental weight decrease for older observations
- *CVFDT (Concept-adapting very fast decision trees)*

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Two examples for concept drift detection algorithms
Informed methods can be distribution- or error rate-based methods

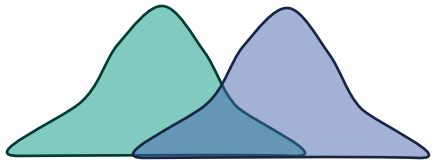
	Data distribution-based drift detection	Error rate-based drift detection
Goal	Quantify **dissimilarity between distribution** of old data and new data with distance function	Detect changes by considering the error rate of the **underlying machine learning model**
Concept		
Example Algorithms	Possible distances: Kullback-Leibler divergence or Kolmogorow-Smirnow	Next! ADWIN, Page-Hinkley-Test
In practice	Consistency checks for input data (mean, variance) Next!	Often manual supervision of prediction quality and impact on KPIs
Disadvantage	Computationally intensive	Requires true labels for drift detection

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Two examples for concept drift detection algorithms
Cost functions evaluate jumps in distribution characteristics

Data distribution-based drift detection

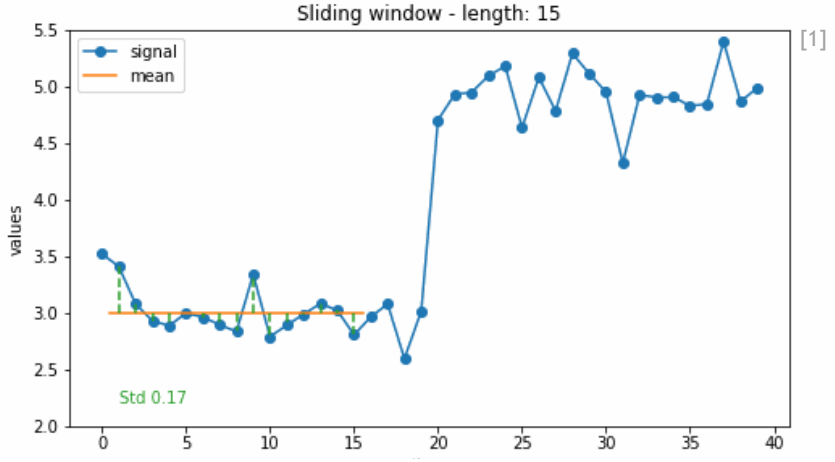
Concept



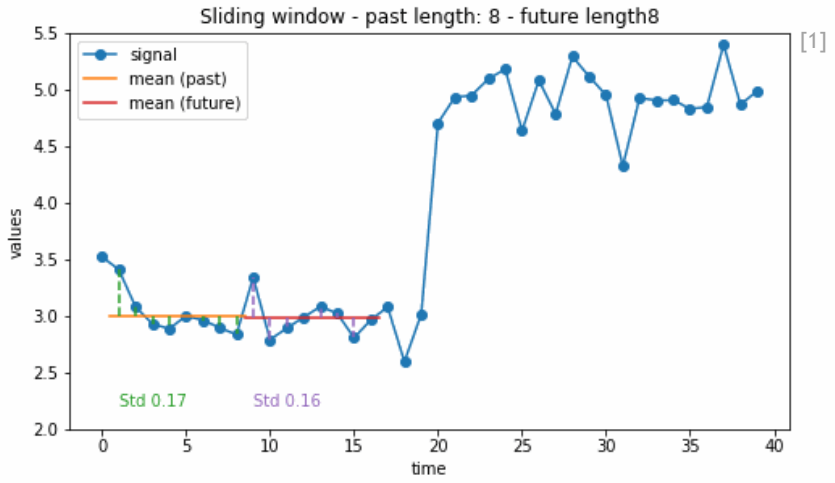
Cost functions – a simple example

- Change point detection uses a **sliding window** with a cost function to identify changes in the signal
- Standard deviation can **detect changes in mean**, rising when the **signal jumps**.
- Change points are detected via **(a)** comparison to a fixed threshold or **(b)** comparing a second sliding windows

(a) Change points are marked if costs **exceed a threshold** in std.



(b) change point can be detected by **comparing the costs** of these two windows



Source: Gama et al. (2014), A Survey on Concept Drift Adaptation
[1] https://www.iese.fraunhofer.de/blog/change-point-detection/

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Two examples for concept drift detection algorithms
Adaptive Sliding Window (ADWIN) compares sliding windows

Adaptive Sliding Window (ADWIN) Algorithm

ADWIN: ADAPTIVE WINDOWING ALGORITHM

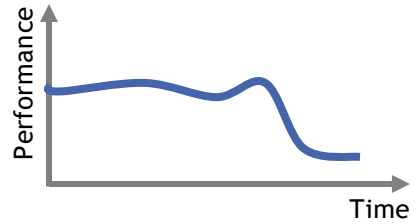
```
1  Initialize Window W
2  for each t > 0
3        do W ← W ∪ {x_t} (i.e., add x_t to the head of W)
4           repeat Drop elements from the tail of W
5              until |μ̂_{W_0} − μ̂_{W_1}| ≥ ε_cut holds
6                 for every split of W into W = W_0 · W_1
7           output μ̂_W
```

Figure 1: Algorithm ADWIN.

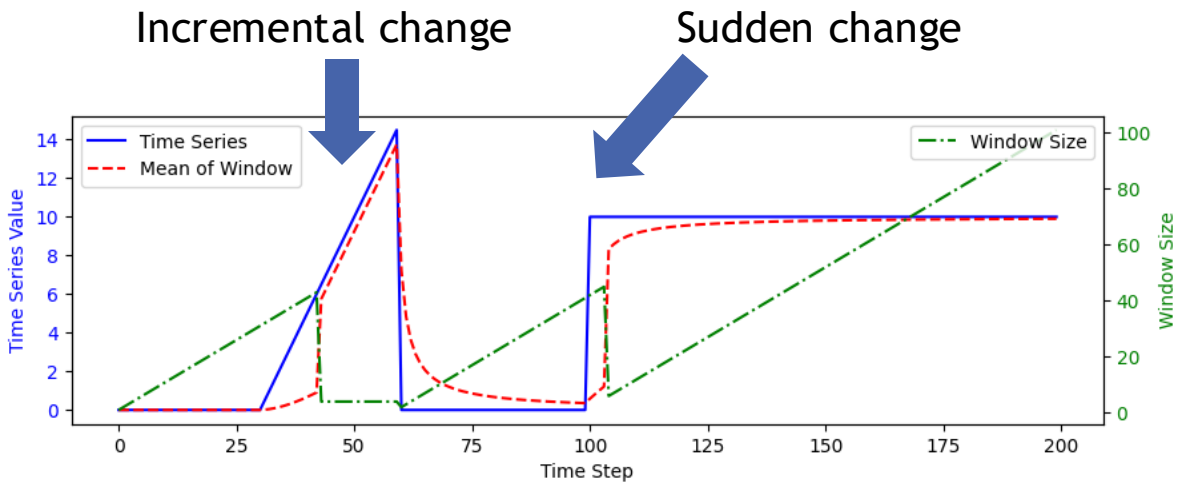
Next!

- Uses a detection window W which iteratively adapts
- Whenever two large (sub)windows of W exhibit distinct enough means, algorithm drops older elements
- Threshold ε_{cut} defined by Hoeffding bound

Error rate-based drift detection



Concept

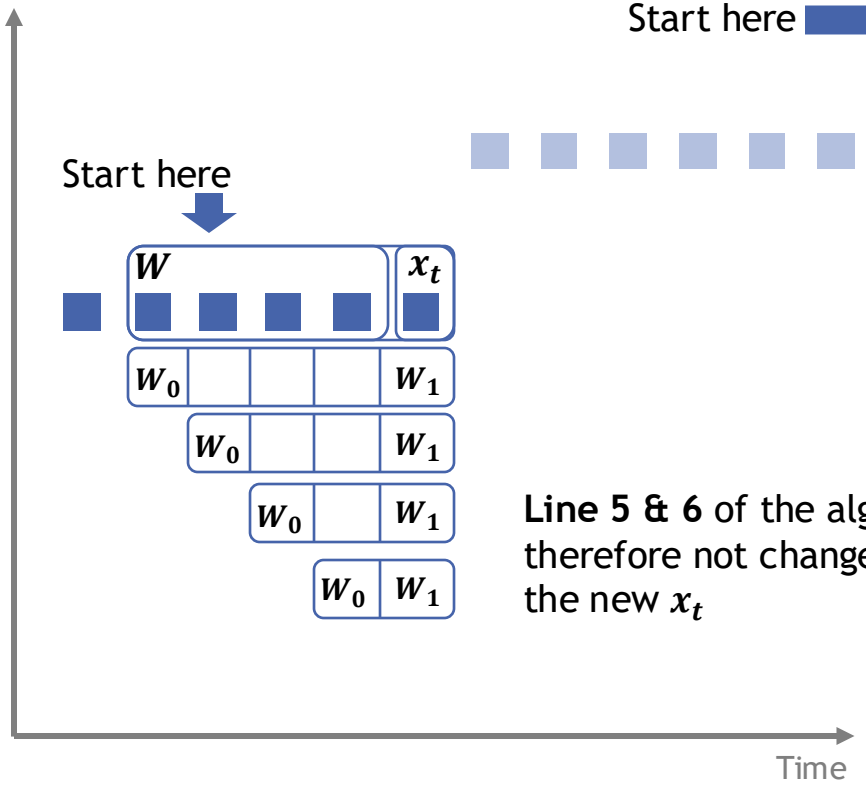


Bifet, Albert, and Ricard Gavalda. "Learning from time-changing data with adaptive windowing." SIAM 2007.

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Two examples for concept drift detection algorithms
Adaptive Sliding Window (ADWIN) compares sliding windows

Sudden changes



ADWIN: ADAPTIVE WINDOWING ALGORITHM

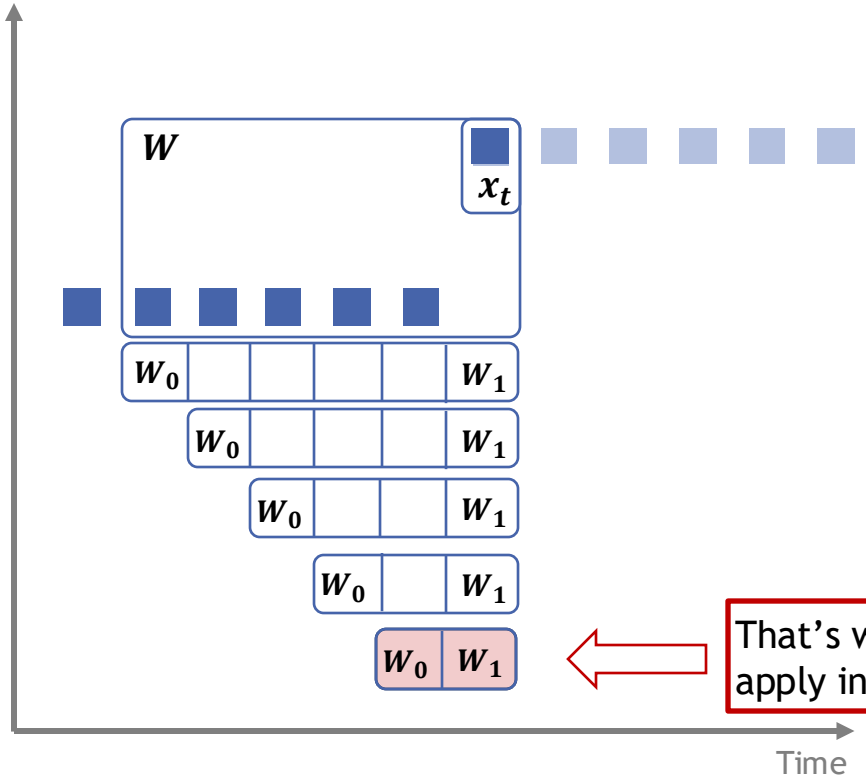
1 Initialize Window W
2 **for** each $t > 0$
3 **do** $W \leftarrow W \cup \{x_t\}$ (i.e., add x_t to the head of W)
4 **repeat** Drop elements from the tail of W
5 **until** $|\hat{\mu}_{W_0} - \hat{\mu}_{W_1}| \geq \epsilon_{cut}$ holds
6 for every split of W into $W = W_0 \cdot W_1$
7 output $\hat{\mu}_W$

Figure 1: Algorithm ADWIN.

Line 5 & 6 of the algorithm never applied, therefore not change is detected and W grows by the new x_t

Two examples for concept drift detection algorithms
Adaptive Sliding Window (ADWIN) compares sliding windows

Sudden changes



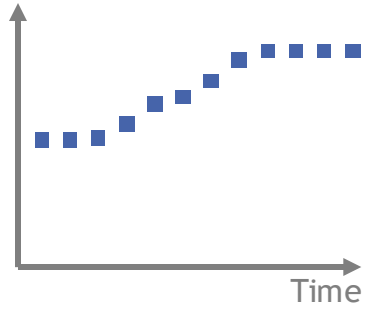
ADWIN: Adaptive Windowing Algorithm

1 Initialize Window W
2 **for** each $t > 0$
3 **do** $W \leftarrow W \cup \{x_t\}$ (i.e., add x_t to the head of W)
4 **repeat** Drop elements from the tail of W
5 **until** $|\hat{\mu}_{W_0} - \hat{\mu}_{W_1}| \geq \epsilon_{cut}$ holds
6 for every split of W into $W = W_0 \cdot W_1$
7 output $\hat{\mu}_W$

Figure 1: Algorithm ADWIN.

That's where **line 5 & 6** of the algorithm apply in the case of **sudden change**

Slow changes



How does ADWIN behave in the case of slow change?



1. Introduction
2. Theoretical foundations of concept drift & adaption mechanism
3. Two examples for concept drift detection algorithms
4. **Real-world examples for concept drift handling**

Real-world examples for concept drift handling
Various threats exists for validity of machine learning microservices

Trigger	Development vs Deployment	Environment of the service	Application of the service
Explanation	• During development, models are often trained with well-defined and curated data sets • In deployment, data input shows great variance • **Example:** Outliers in input data during deployment	• Change in the environment of the service with corresponding data changes • Influence on prediction quality • **Example:** Sensor of a production machine wears out	• Application of service affects prediction quality • **Examples:** • Continuous predictive policing service • Changing user behavior due to machine learning services (different click stream)

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Real-world examples for concept drift handling

Studies on concept drift have been performed in various domains

	Monitoring & control	Information management	Analytics & diagnostics
Drift type	• Real time monitoring • Typically fast concept drift	• Time-stamped data • Sudden or gradual concept drift	• Time-stamped data • Slower concept drift (e.g. population drift)
Use cases	• **Monitoring for Management:** Monitoring output quality in chemical production • **Automated control:** Soccer playing robots • **Anomaly detection:** Network intrusion detection	• **Personal assistance:** News categorization, spam filtering • **Marketing:** customer segmentation for cars, recommender systems • **Management:** Archiving of documents	• **Forecasting:** Macroeconomic forecasts • **Medicine:** Antibiotic resistance • **Security:** Biometric authentication

Source: Zliobaite et al. (2016), An Overview of Concept Drift Applications

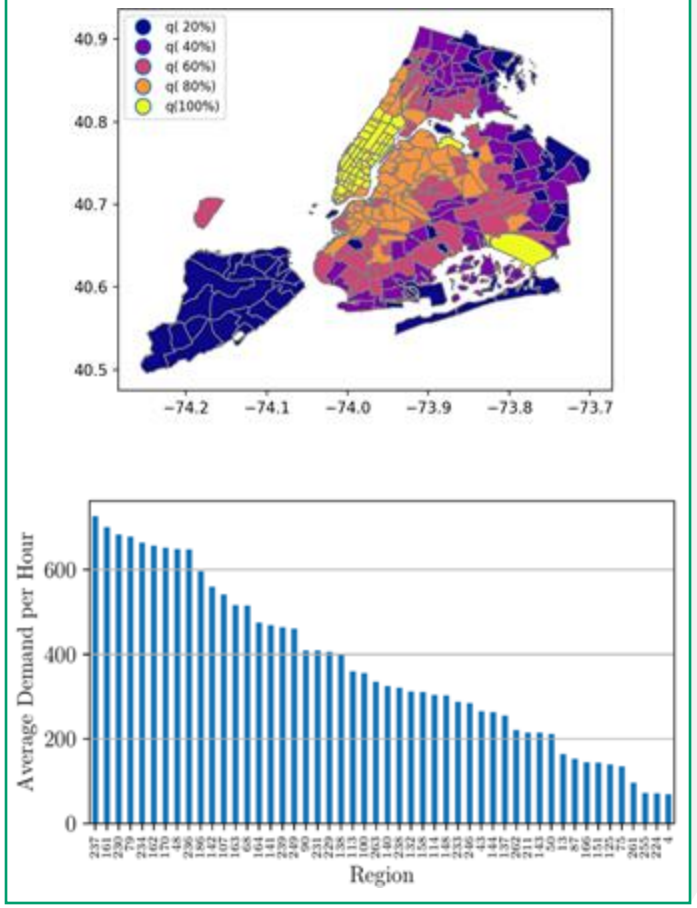
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Real-world examples for concept drift handling
Exemplary drifts in NYC taxi demand

Time series examples with sudden drift:

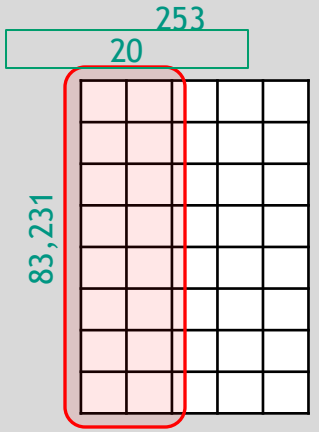
- Temperature
- Stock prices
- Taxi demand
- Demand for groceries

New York City Taxi data set



- ~1.4 billion rides (yellow cabs)
- 01/2009 – 06/2018
- Features: information regarding pickup and drop-off, fare and tip
- Various preprocessing steps

Objective: Predict hourly taxi demand per taxi zone



~ 21 million observations

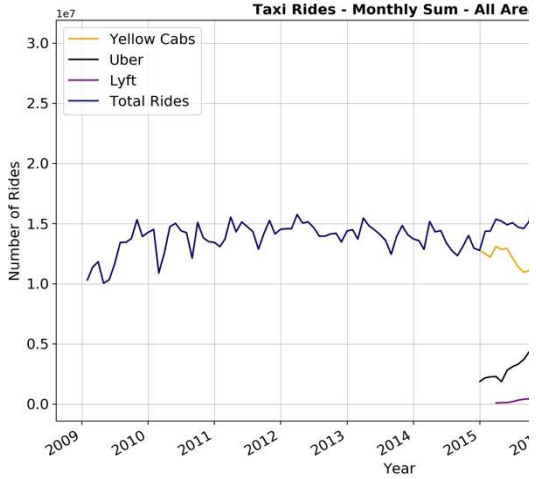
Source: Baier, Hofmann, Kühl, Mohr, Satzger (2020), Handling Concept Drifts in Regression Problems – the Error Intersection Approach

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Real-world examples for concept drift handling
Exemplary drifts in NYC taxi demand

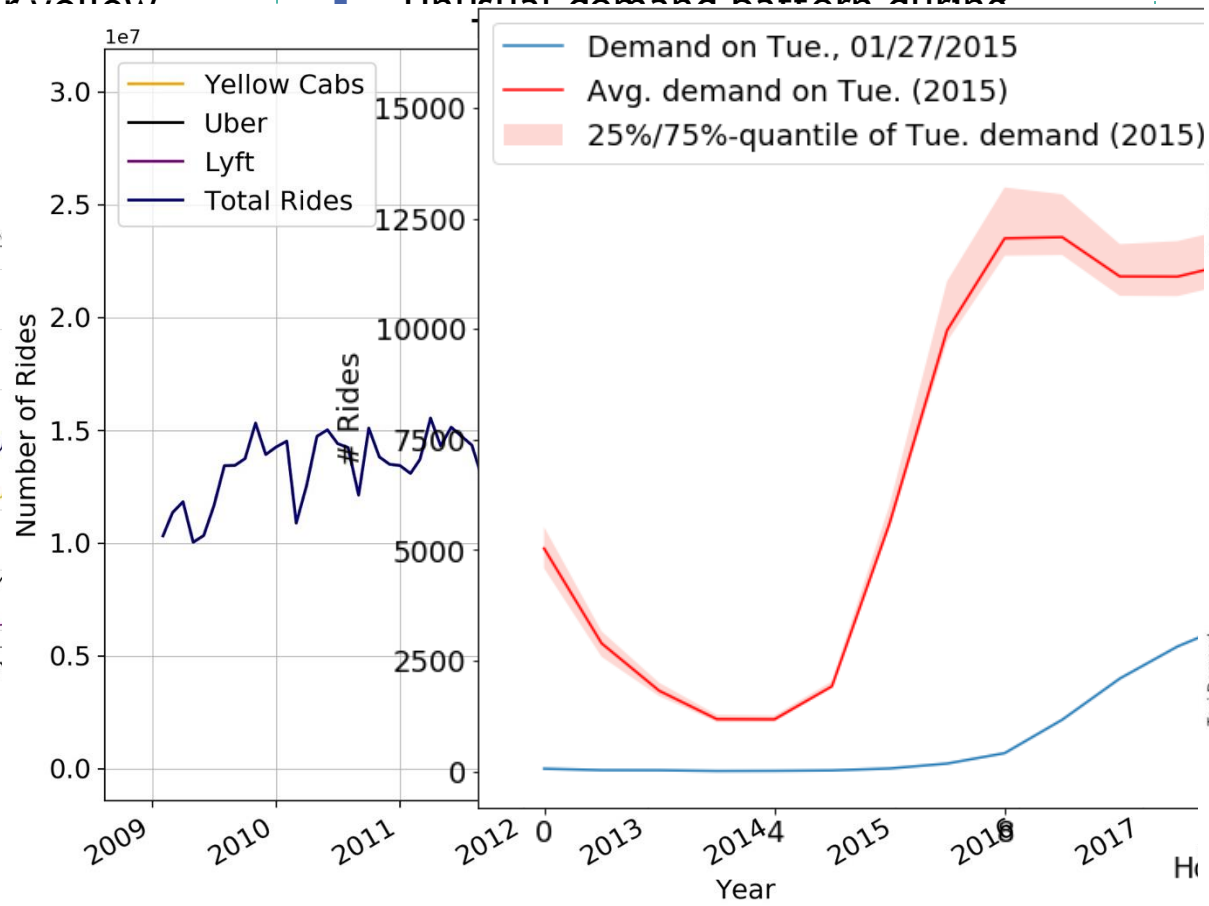
Incremental drift

- Decreasing demand for yellow cabs over time
- New competitors

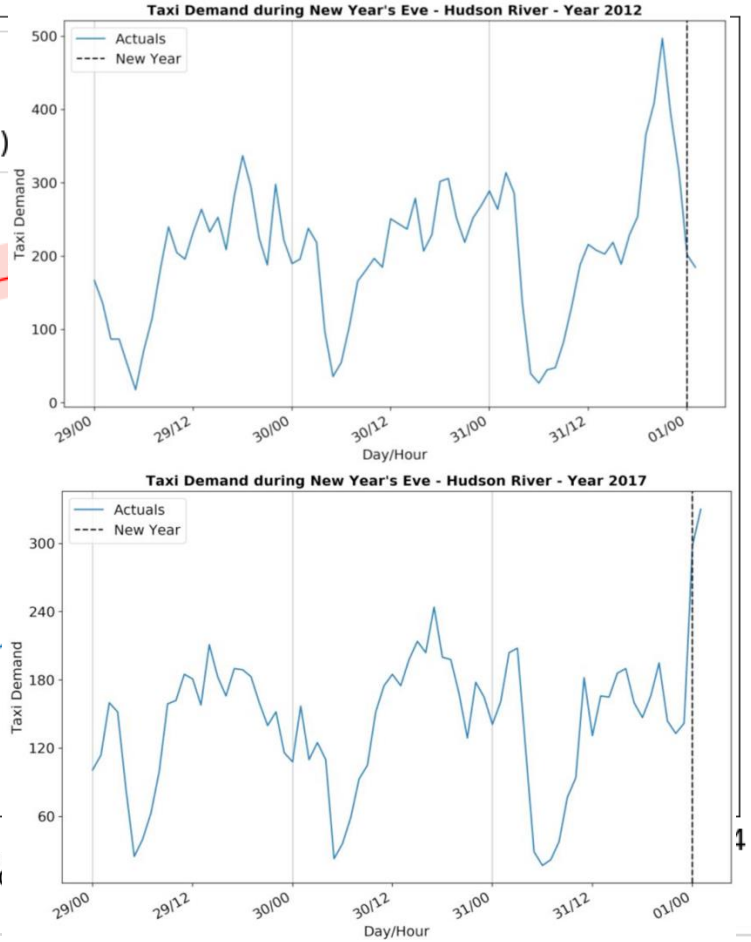


Sudden drift

- Unusual demand pattern during



Seasonal drift

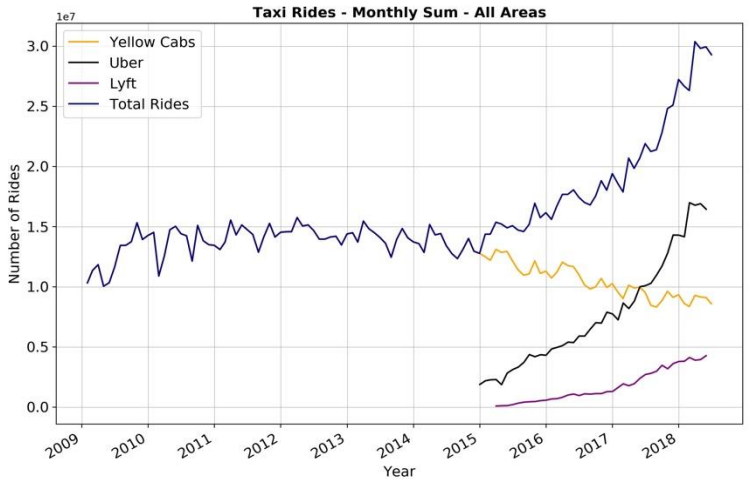


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Real-world examples for concept drift handling
Exemplary drifts in NYC taxi demand

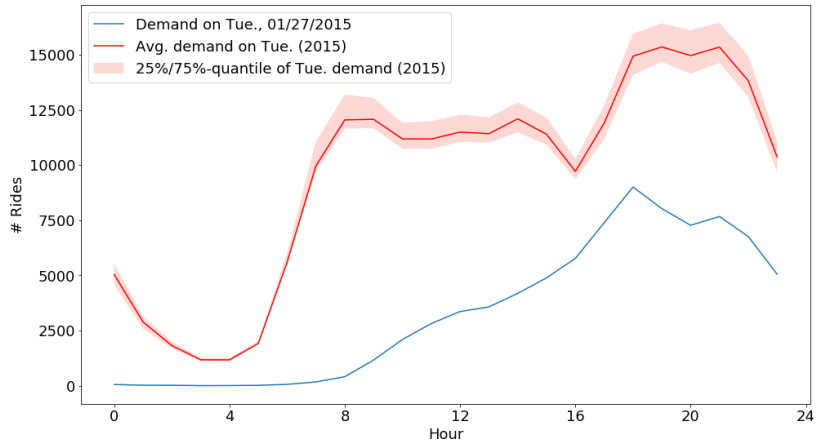
Incremental drift

- Decreasing demand for yellow cabs over time
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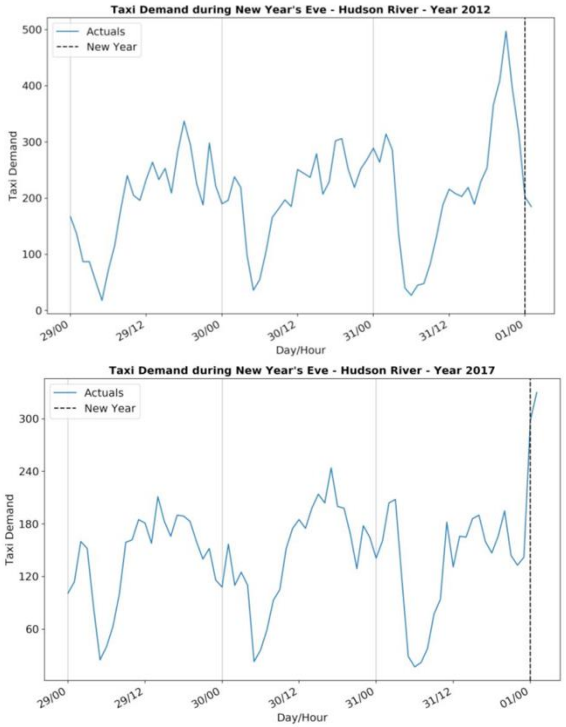
Sudden drift

- Unusual demand pattern during extreme weather phenomena such as blizzards



Seasonal drift

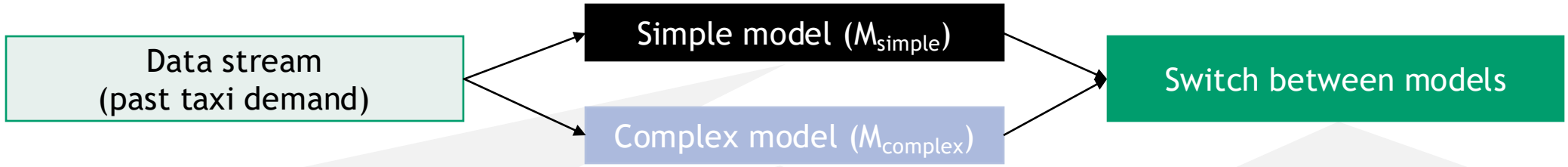
- Public holidays



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Real-world examples for concept drift handling
Switch between models to handling concept drift

Error Intersection Approach: Switch to simple model in case of sudden drift and use complex model for ordinary days



Simple model (M_{simple})

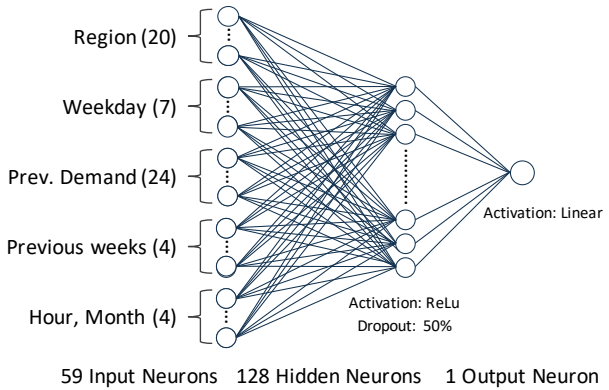
Simple model needs to quickly adapt to changing circumstances

Random walk model:

$$\hat{y}_{s,t,r} = y_{t-1,r}$$

s: simple, r: region

Complex model ($M_{complex}$)



Switching mechanism

$$Err_{M,t,r} = \left|\hat{y}_{M,t,r} - y_{t,r}\right|$$

$$EWMA_{M,t,r} = \begin{cases} Err_{M,1,r}, & t = 1 \\ \alpha * Err_{M,t,r} + (1-\alpha) * EWMA_{M,t-1,r}, & t > 1 \end{cases}$$

$$\alpha = \frac{2}{N+1}, \quad N = 6$$

$$E_{M,t} = \frac{1}{R}\sum_{r=1}^{R} EWMA_{M,t,r}$$

$$M \in \{s(imple), c(omplex)\}$$

$$choose \ \underset{s,c}{\operatorname{argmin}}(E_{s,t-1}, E_{c,t-1})$$

r: region, N: parameter

Source: Baier, Hofmann, Kühl, Mohr, Satzger (2020), Handling Concept Drifts in Regression Problems - the Error Intersection Approach

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Real-world examples for concept drift handling
EIA performs well with unusual demand patterns

Date	Abs. RMSE improvement	Rel. RMSE improvement	# Predictions by M_{simple}	Probable Drift Cause
2012-07-04	5.07	3.96%	14/24	4th of July
2012-10-29	24.41	31.02%	22/24	Hurricane Sandy
2012-12-25	**9.22**	**12.61%**	**17/24**	**Christmas Day**
2013-08-01	9.35	8.61%	10/24	? (unknown)
…	…	…	…	…
2016-01-23	**66.61**	**61.14%**	**16/24**	**Blizzard**
2017-06-25	5.48	10.12%	10/24	? (unknown)
2018-03-21	15.21	17.43%	14/24	Cyclone (Nor'easter)

Table: Excerpt of days with frequent use of M_{simple}

- We also find evidence for days where we cannot explain the drift cause
- Influencing factors are easy to identify from hindsight and might not be obvious in real time

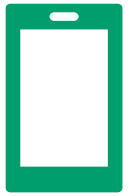
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Summary



Constant Monitoring

A **constant monitoring of deployed machine** learning solutions is crucial for ensuring their proper functionality / validity over time



Application domain

Concept drift is a phenomenon which can be observed in nearly **all application domains**



Implementation

The **change detection and adaptation options are manifold** and different choices must be made during the implementation



Use-case specific

Real-world solutions for machine learning services usually need to be **tailored to the specific use case**

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